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Activation of CBASS Cap5 endonuclease immune effector by cyclic nucleotides

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The bacterial cyclic oligonucleotide-based antiphage signaling system (CBASS) is similar to the cGAS–STING system in humans, containing an enzyme that synthesizes a cyclic nucleotide on viral infection and an effector that senses the second messenger for the antiviral response. Cap5, containing a SAVED domain coupled to an HNH DNA endonuclease domain, is the most abundant CBASS effector, yet the mechanism by which it becomes activated for cell killing remains unknown. We present here high-resolution structures of full-length Cap5 from *Pseudomonas syringae* (*Ps*) with second messengers. The key to *Ps*Cap5 activation is a dimer-to-tetramer transition, whereby the binding of second messenger to dimer triggers an open-to-closed transformation of the SAVED domains, furnishing a surface for assembly of the tetramer. This movement propagates to the HNH domains, juxtaposing and converting two HNH domains into states for DNA destruction. These results show how Cap5 effects bacterial cell suicide and we provide proof-in-principle data that the CBASS can be extrinsically activated to limit bacterial infections.

In humans, the cGAS–STING pathway is the cell autonomous immune system for defense against viruses and other pathogens[1]. Bacteria also encode immune systems for defense against viruses (phages) with many new ones discovered only recently[2,3]. These include the widespread CBASS with structural and functional similarities to the cGAS–STING system in humans[4,5]. Both systems use an enzyme that senses viral infection and synthesizes a cyclic nucleotide second messenger that then binds an effector protein to evoke the antiviral response[2,3,5]. The CBASS is specifically defined by a cGAS/DncV-like nucleotidyltransferase cyclase (CD-NTase) enzyme and one or more CD-NTase-associated protein (Cap) effectors[5,6]. The CBASS effectors do not activate transcription-like STING, but instead invoke death of the infected cell (or abortive infection) to prevent propagation of the virus within the bacterial cell population[7–14].

CBASS effectors can sense a cyclic nucleotide produced by the cognate CD-NTase via an animal-like STING domain, or far more commonly by a structurally and functionally related SAVED domain[6,9–12,15]. The SAVED domain is a fusion of two CARF modules co-opted from the type III CRISPR–Cas system[9]. The largest group of these SAVED domain containing proteins are Cap5 HNH endonucleases, which once activated, degrade genomic DNA resulting in abortive infection[9]. Despite their abundance, the mechanism by which the Cap5 effectors become activated by a second messenger for abortive infection remains unknown. Crystallographic studies of full-length Cap5 from *Lactoccossus lactis* (*Ll*) and *Asitccacaulis* sp. (*As*) without a ligand and the structure of ligand-bound *Ll*SAVED domain have been reported recently[10]. However, the all-important structure of a full-length Cap5 in complex with a second messenger (ligand) for understanding the activation mechanism has remained elusive.

We present here high-resolution structures of full-length Cap5 from *Pseudomonas syringae* (*Ps*) in complex with second messengers. We show that the key to *Ps*Cap5 activation by a second messenger is an unexpected dimer to tetramer transition that catalytically poises the HNH endonuclease domains for DNA destruction.

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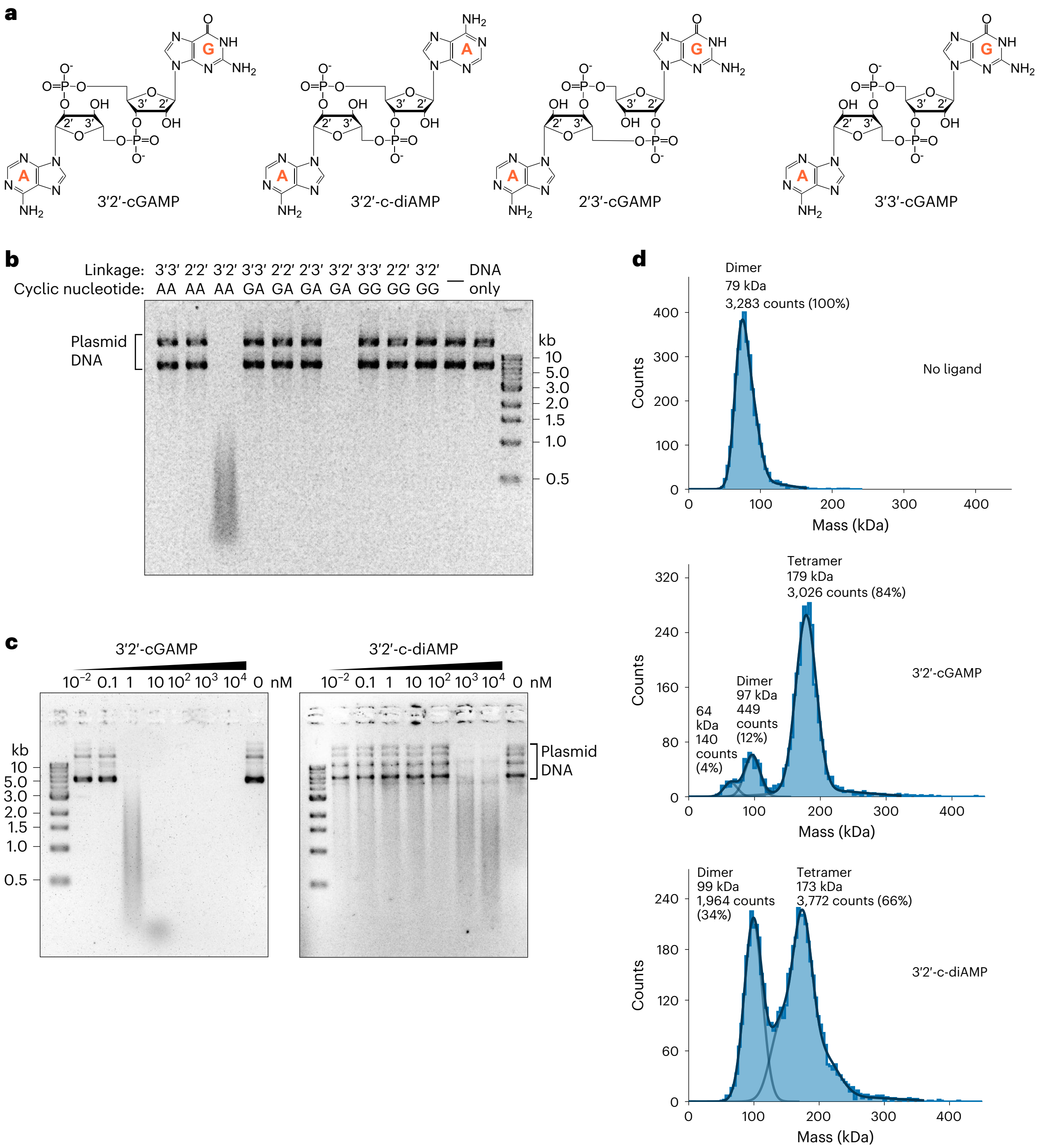

Fig. 1 | Identification of activating ligands and the oligomerization states of *Ps*Cap5. a, Chemical structures of 3′2′-cGAMP, 3′2′-c-diAMP, eukaryotic STING ligand 2′3′-cGAMP and bacterial canonically 3′3′-linked 3′3′-cGAMP. **b**, Plasmid DNA digestion by *Ps*Cap5 in the presence of cyclic purine dinucleotides with 3′3′, 2′3′ and 2′2′ linkages using 500 nM protein and ligand concentrations. Plasmid digestion is observed with only 3′2′-cGAMP (3′2′GA) and 3′2′-c-diAMP (3′2′AA). Plasmid DNA digestion with other di-, tri- and tetra-nucleotides is shown in Extended Data Fig. 1a. **c**, Determination of the *Ps*Cap5-activating concentrations for 3′2′-cGAMP and 3′2′-c-diAMP. Concentration of *Ps*Cap5 is 50 nM and the ligand concentrations are varied in the 10^{-2}–10^{4} nM range as indicated. Each gel is a representative of at least three independent experiments. **d**, Mass photometry analysis of apo *Ps*Cap5 and in presence of 3′2′-cGAMP and 3′2′-c-diAMP. Solid lines represent the Gaussian function fit to the main species observed on particle counts versus molecular mass distribution histograms. The expected theoretical masses of apo *Ps*Cap5 monomer, dimer and tetramer are 42.7, 85.4 and 170.8 kDa, respectively.

Results

Identification of *Ps*Cap5-activating ligand

Bacterial CBASS CD-NTases can use all four nucleotides for synthesizing cyclic di- and tri-nucleotide second messengers containing canonical 3′–3′ and noncanonical 2′–3′ phosphodiester bonds[9,16]. To identify a putative activating ligand for *Ps*Cap5, we screened a series of commercially available canonical bacterial 3′–5′/3′–5′ (3′3′) linked cyclic di-, tri- and tetra-nucleotides as well as mixed-linked 2′–5′/3′–5′ (2′3′) (or 3′–5′/2′–5′ (3′2′) linked depending on which base is named first) and 2′–5′/2′–5′ (2′2′) cyclic purine dinucleotides for their ability to activate *Ps*Cap5 in a plasmid DNA degradation assay (Fig. 1a,b and Extended Data Fig. 1a). We observed plasmid digestion with only two of the tested ligands: 3′2′-cGAMP (3′2′GA) and 3′2′-c-diAMP (3′2′AA). Addition of 3′2′-cGAMP in particular leads to complete plasmid degradation whose products are too short to be detected. Notably, the animal STING ligand 2′3′-cGAMP does not activate *Ps*Cap5 (Fig. 1b). Likewise, we do not observe DNA cleavage with the symmetrically linked canonical bacterial 3′3′-cGAMP second messenger.

Table 1 | Data collection and refinement statistics (molecular replacement)

	*Ps*Cap5 with 3′2′-cGAMP (two tetramers) (PDB 8FMH)	*Ps*Cap5 with 3′2′-c-diAMP (three tetramers) (PDB 8FMG)	*Ps*Cap5 with 3′2′-c-diAMP (one tetramer) (PDB 8FMF)	*Ps*Cap5 with no ligand (Apo) (six dimers) (PDB 8FM1)
Data collection				
Space group	$P2_1$	$C2$	$P3_22$	$P4_12_12$
Cell dimensions				
a, *b*, *c* (Å)	67.97, 295.68, 84.05	145.19, 80.52, 406.84	83.54, 83.54, 401.96	159.43, 159.43, 433.88
α, *β*, *γ* (°)	90.0, 113.8, 90.0	90.0, 90.6, 90.0	90.0, 90.0, 120.0	90.0, 90.0, 90.0
Resolution (Å)	147.8–1.87 (2.11–1.87)[a]	203.4–1.79 (2.00–1.79)	134.0–2.11 (2.32–2.11)	112.7–3.16 (3.34–3.16)
R_{merge}	0.125 (0.753)	0.071 (0.645)	0.251 (1.961)	0.087 (1.456)
$I / \sigma I$	6.9 (1.9)	7.0 (1.6)	8.8 (1.6)	12.8 (1.6)
Completeness (%)	93.6 (67.7)	88.7 (63.2)	94.8 (81.4)	95.4 (51.1)
Redundancy	4.0 (3.9)	2.1 (2.8)	12.0 (14.3)	6.8 (7.5)
Refinement				
Resolution (Å)	38.45–1.87	58.12–1.79	53.78–2.11	112.7–3.16
No. reflections	151,489	267,447	64,174	85,838
R_{work} / R_{free}	0.210 / 0.253	0.179 / 0.225	0.190 / 0.237	0.250 / 0.299
No. atoms	23,394	36,515	11,857	32,962
Protein	22,017	33,481	11,124	32,950
Ligand	383	528	269	—
Ion	18	25	11	12
Water	976	2481	453	—
B factors				
Protein	54.04	33.88	40.66	121.58
Ligand	43.16	30.86	47.56	—
Ion	66.76	34.32	40.09	110.00
Water	46.07	33.02	34.41	—
R.m.s. deviations				
Bond lengths (Å)	0.003	0.004	0.003	0.006
Bond angles (°)	0.62	0.70	0.56	1.05

[a]Values in parentheses are for highest-resolution shell.

To better assess the concentrations of the activating ligands required for DNA digestion, we lowered *Ps*Cap5 protein concentration to 50 nM and varied the ligand concentrations in the 10^{-2}–10^{4} nM range (Fig. 1c). We detected DNA cleavage with the strongly activating 3′2′-cGAMP ligand at a concentration of 1 nM, compared with a roughly 1,000-fold higher concentration of 1 µM with the weakly activating 3′2′-c-diAMP dinucleotide. This highlights the remarkable specificity of *Ps*Cap5 for the 3′2′-cGAMP second messenger compared to the closely related 3′2′-c-diAMP. The only difference between 3′2′-cGAMP and 3′2′-c-diAMP is the substitution of guanine for adenine at one of the nucleobase positions. Our results are in line with DNA digestion studies performed with other SAVED domain containing effectors, which are similarly activated by low nM concentrations of cognate ligands. For example, several Cap4 proteins display robust DNA cleavage activities with 50 nM of cyclic tri-nucleotides[9]. Similarly, *B. pseudomallei* Cap5 is active with 50 nM of 3′3′-cGAMP (ref. 9) and *As*Cap5 and *Ll*Cap5 proteins are activated by 1–10 nM 3′2′-cGAMP (ref. 10).

Oligomeric states of *Ps*Cap5

To assess the oligomeric states of *Ps*Cap5 in absence and presence of the activating ligands, we used mass photometry. The characterization of apo *Ps*Cap5 revealed that the protein exists predominantly as a dimer in solution, wherein 100% of the detected particles belonged to a single peak with an estimated molecular weight of 79 kDa, which is approximately double the theoretical molecular weight of 42.7 kDa for the monomer (Fig. 1d, top panel). On the addition of 3′2′-cGAMP the mass distribution showed two peaks, corresponding to a dimer and a tetramer (Fig. 1d, middle panel). Addition of the weaker activating dinucleotide 3′2′-c-diAMP also lead to the tetramer formation (Fig. 1d, bottom panel). We did not detect any higher-order oligomers even with long 3 h incubation times. These data strongly indicate that the activated ligand-bound form of *Ps*Cap5 is a tetramer.

Activated ligand-bound *Ps*Cap5 tetramer

We solved the structures of *Ps*Cap5 with its favored ligand 3′2′-cGAMP, as well as the less preferred 3′2′-c-diAMP with the highest-resolution structure refined to 1.79 Å (Table 1 and Extended Data Fig. 1b–d). Each *Ps*Cap5 protomer consists of the N-terminal HNH domain (residues 1–96), the C-terminal SAVED domain (residues 127–388) and a mostly α-helical connector (residues 97–126) (Fig. 2a and Extended Data Fig. 2a,b). As expected, the SAVED domain is a fusion of two CARF modules[9] (CARF1 and CARF2) with alternating β-strand and α-helical substructures (Fig. 2b and Extended Data Fig. 3a,b). The protomers assemble into dimer-of-dimers with both 3′2′-cGAMP and 3′2′-c-diAMP. The resulting tetramers (chains A, B, C and D) are similar in all of the

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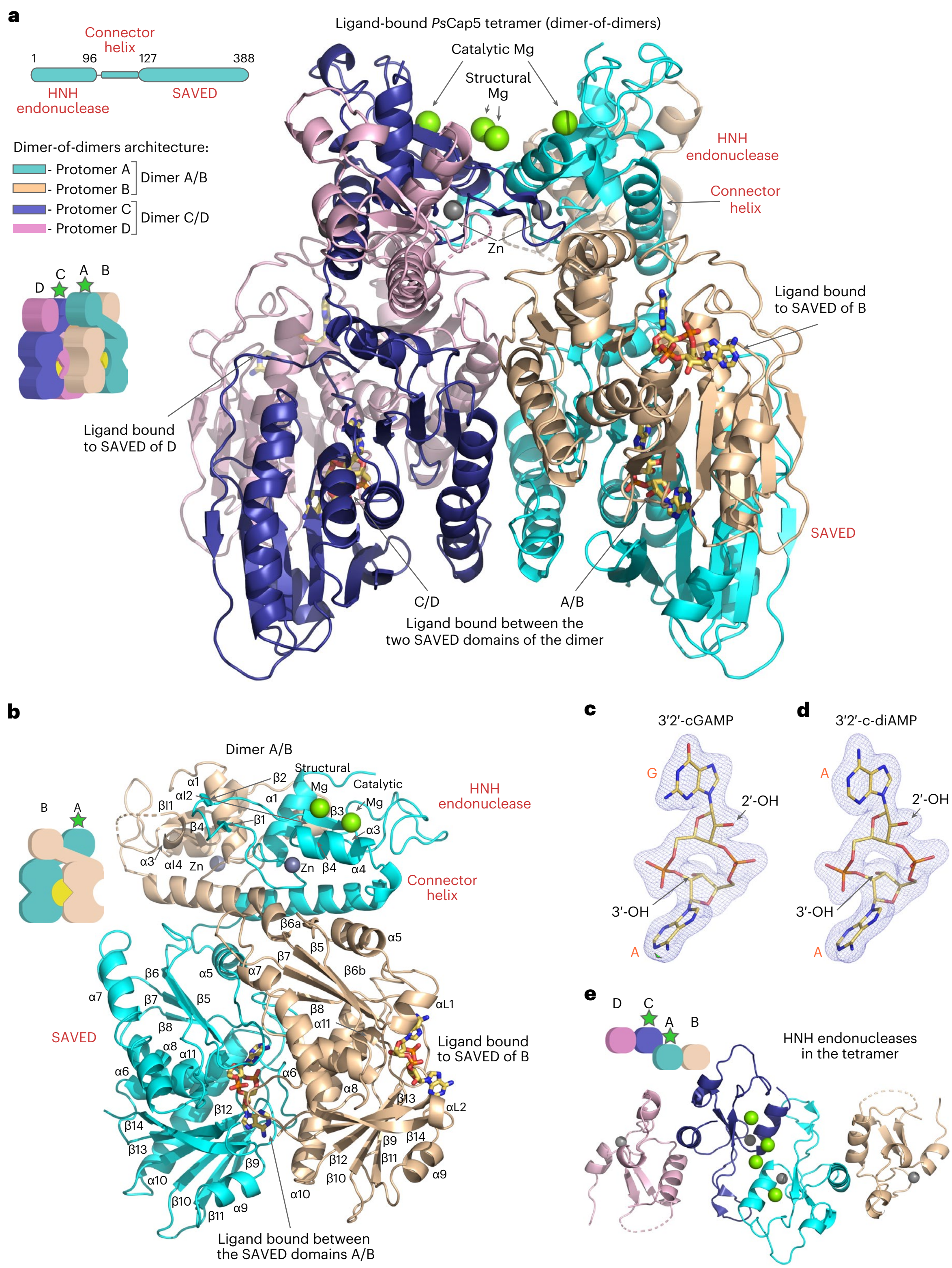

Fig. 2 | Structure of the activated ligand-bound *Ps*Cap5 tetramer. a, The overall dimer-of-dimers architecture of the 3′2′-c-diAMP bound *Ps*Cap5 tetramer. Two crisscross dimers (A/B and C/D), formed by *Ps*Cap5 protomers A and B and C and D, produce an activated tetramer. The HNH domains of protomers A and C are in the catalytically active conformation noted by a green star. There are two 3′2′-c-diAMP ligands bound to the primary sites between the SAVED domains of dimers A/B and C/D, respectively, and to the secondary sites on the outside of the SAVED of B and D, respectively. **b**, The crisscross dimer A/B with secondary structure elements labeled. The SAVED domains of A and B present different surfaces for interactions with the ligand; see Extended Data Fig. 3 for details. **c**,**d**, Polder electron density maps contoured at the 4σ level are shown as blue mesh at 1.87 Å resolution for 3′2′-cGAMP (**c**) and at 1.79 Å resolution for 3′2′-c-diAMP ligands (**d**). **e**, HNH endonuclease domains arrangement in the tetramer. The HNH endonucleases of protomers A and C are positioned next to each other in a dimeric catalytically active state, whereas the HNH domains of protomers B and D are on the outside of the assembly in a catalytically inactive state.

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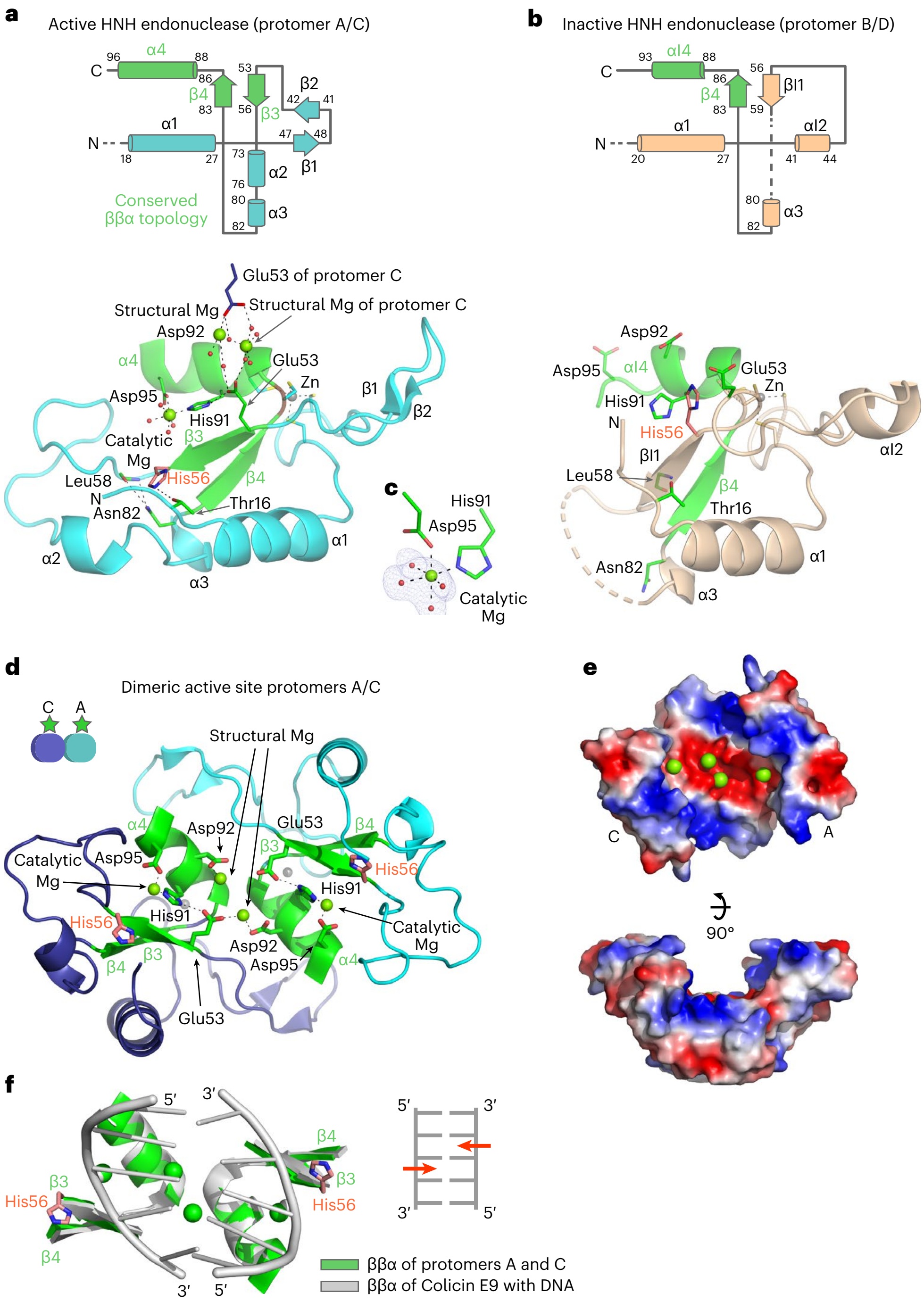

Fig. 3 | Active and inactive HNH endonuclease domains of *Ps*Cap5 tetramer and the catalytically competent dimeric active site. a, Topology and structure of the active conformation of the HNH endonuclease (protomers A and C) with the characteristic ββα-topology of HNH endonucleases colored in green. Catalytic His56 is highlighted in red. **b**, Topology and structure of the inactive conformation of the HNH endonuclease (protomers B and D). **c**, Polder electron density map for catalytic Mg^{2+} ion of protomer A at 1.79 Å resolution contoured at the 5σ level. **d**, Structure of the dimeric active site formed by the HNH endonucleases of protomers A and C of the tetramer. **e**, Electrostatic surface representation of the dimeric active site with views from the top and the side. **f**, Model of DNA in the dimeric active site of *Ps*Cap5 tetramer based on Colicin E9–DNA complex (PDB ID 1V15; ref. 22).

ligand-bound structures (Fig. 2a and Extended Data Fig. 1b–d), and we note differences as we proceed.

*Ps*Cap5 dimers are arranged in a crisscross fashion to form the activated tetramer, wherein protomers A/B comprise one dimer and protomers C/D comprise the other dimer (Fig. 2a). The primary ligand binding pocket is formed mainly by the SAVED domain of protomer A (or C), with the SAVED domain of protomer B (or D) forming a lid over the pocket (Fig. 2b and Extended Data Figs. 2b and 3a,b). Thus, the two protomers of the A/B dimer (or C/D) interact asymmetrically with the ligand (Fig. 2a), which is exceptionally well defined (Fig. 2c,d). There is a secondary ligand binding site on the outside of SAVED domains of protomers B and D (Fig. 2a,b) that exploit the same surface involved in ligand binding by the SAVED domains of A and C.

The HNH endonuclease domains of protomers A and C are in a catalytically active state, while the HNH domains of protomers B and D are in a catalytically inactive state (Figs. 2e and 3a,b). In particular, the secondary structure of the inactive protomers (Fig. 3b) deviates majorly from the characteristic ββα-topology of HNH endonucleases[17]. Thus, whereas residues 53–56 in the catalytically active HNH domains comprise the first β-strand (β3) of the ββα substructure and carry the all-important catalytic His56, the same residues in the inactive domains form a loop and it is the ensuing residues (56–59, which we label as βI1) that form the first β-strand of the ββα substructure (Fig. 3a,b). The net effect of this structural rearrangement is that catalytic His56 is displaced by as much as around 7.5 Å (Cα to Cα distance) from the position in the catalytically active HNH domains (Fig. 3a,b), accompanied by other changes described below.

Activation of HNH endonuclease

The two SAVED domains in each dimer (A/B or C/D) are brought in close proximity by the binding of the ligand between them. This 'closed' conformation of SAVED domains in turn mediates tetramerization via contacts predominately between protomers A and C (Extended Data Fig. 4a,b). As a consequence, the HNH endonuclease domains of protomers A and C are positioned next to each other in catalytically active states, whereas the HNH domains of protomers B and D on the outside of the tetramer assembly are in catalytically inactive states (Fig. 2e).

Members of HNH endonuclease superfamily share common DNA cleavage and binding motifs, but exhibit very limited sequence homology[17]. The active protomers A and C of *Ps*Cap5 enable the canonical ββα-topology with short antiparallel β3- and β4-strands and α4-helix (Fig. 3a and Extended Data Fig. 2a). His91 and Asp95 of the α4-helix bind a catalytic Mg^{2+} ion with rigid coordination and solvation pattern (Fig. 3c) with the invariant catalytic His56 at the end of β3 functioning as a general base to activate the water nucleophile for phosphodiester bond hydrolysis[17]. Thus, when we mutated His56 to alanine or His91-Asp95 to alanines, the mutant proteins completely lacked enzymatic activity (Extended Data Fig. 5a,b). The β3-strand (residues 53–56) of the active HNH domains is framed by stabilizing interactions that include hydrogen bonds with main chain carbonyl and amide group of Leu58 and the side chain of Asn82 (Fig. 3a); interactions observed in some other HNH endonucleases[17]. A unique feature of the activated HNH domains in our structure is the bridging interactions between them, wherein a structural Mg^{2+} in each domain is coordinated by Asp92 and Glu53 to form an 'ion clasp' (Fig. 3d and Extended Data Fig. 4b). Substitution of Asp92 and Glu53 residues by alanine substantially reduces the DNA cleavage activity of *Ps*Cap5 (Extended Data Fig. 5a,b). Bridging interactions between the active A/C HNH domains are further reinforced by two short antiparallel β-strands (β1 and β2) (Extended Data Fig. 4a).

The inactive HNH domains of protomers B and D lack not only the catalytic Mg^{2+} but also the structural Mg^{2+} (Fig. 3b and Extended Data Fig. 2a). The α-helix equivalent to α4 is shorter in the inactive state (αI4, residues 88–93), positioning the side chain of Asp95 roughly 7 Å away from the putative catalytic Mg^{2+}-coordinating position. In several inactive protomers of the ligand-bound structures reported here, the side chains of Asp95, and of the linker residues Asp97, Asp99, Glu103 and the main chain carbonyl of Tyr101, engage a different Mg^{2+} ion, thus helping to keep Asp95 in the inactive state (Extended Data Fig. 4c).

Mechanism of DNA cleavage

HNH endonucleases can function as monomers and cut one strand of a DNA duplex or as dimers to nick both DNA strands at the same time[17]. The activated HNH domains of protomers A and C of the ligand-bound *Ps*Cap5 tetramer form a crescent-shaped dimeric active site with the α helices of the ββα substructures collocated as in other dimeric HNH endonucleases (Fig. 3d,e and Extended Data Fig. 6). The conserved ββα-elements of the dimer (Fig. 3d) are expected to insert into the minor groove of a DNA duplex (Fig. 3f), as observed with DNA complexes of dimeric HNH endonucleases such as I-PpoI (ref. 18), Hpy99I (ref. 19) and T4 Endo VII (ref. 20) (Extended Data Fig. 6). The distance between the two catalytic Mg^{2+} ions (19.3 Å) is similar to that observed in the I-PpoI and Hpy99I DNA complexes (18.7 and 22.3 Å, respectively) and consistent with the diameter of B-DNA (~20 Å) for cleavage of opposite strands. In I-PpoI, Hpy99I and T4 Endo VII, the dimeric active site is positively charged as would be expected for interactions with a negatively charged sugar-phosphate DNA backbone. By contrast, the surface area between the catalytic ions in *Ps*Cap5 is negatively charged to coordinate the structural Mg^{2+} ions of the 'ion clasp'. These ions may directly contact the DNA backbone and appear to be surrogates for the Lys and Arg residues in other HNH endonucleases. Although the HNH domains of protomers B and D are catalytically inactive and devoid of Mg^{2+} ions, a substantial portion of their surface is positively charged and may contribute to DNA binding (Extended Data Fig. 6).

Recognition of 3′2′-cGAMP and 3′2′-c-diAMP second messengers

How do we explain the specificity of *Ps*Cap5 for 3′2′-cGAMP over 3′2′-c-diAMP? From the structure of the complex, specificity for the G nucleobase is achieved via interactions with residues from the SAVED of protomer A only (Fig. 4a). The side chain of Arg366 recognizes the Hoogsteen edge of the G via contacts with the N7 and O^6 atoms and the phenolic OH-group of Tyr304 contacts the N^2 amino group of the base. The Watson–Crick edge of the G is hydrated with three water molecules and the water bridge extends to the main chain carbonyl of Gln279. Recognition of the A nucleobase is achieved via residues from both A and B protomers. Residues Asp231 and Arg232 of protomer B make direct hydrogen bonds with the Hoogsteen edge of the A, whereas Phe240 and Ala343 of protomer A make water-mediated contacts with the A. Specificity for the 3′2′ (3′–5′/2′–5′) linkage is derived through a combination of direct and water-mediated hydrogen bonds from residues His138, Arg178, Ile219, Arg242, Ser277, Tyr304, Ile341, Ala343 and Asp365 of protomer A and residues Gln356 and His357 of protomer B. In comparison to the *Ps*Cap5–3′2′-cGAMP complex, the nucleobase A in place of the G in the *Ps*Cap5–3′2′-c-diAMP complex loses two critical hydrogen bonds: one from the side chain of Arg366 (to O^6 of G) and another from the side chain of Tyr304 (to N^2 of G) (Fig. 4b). The retained hydrogen bonds are from Arg366 to the N7 atom and the water bridge to Gln279. The loss of these hydrogen bonds helps to explain the lower preference of *Ps*Cap5 for 3′2′-c-diAMP as the second messenger. Contacts to the 3′–5′ and 2′–5′ sugar-phosphate linkages are nearly identical in the *Ps*Cap5–3′2′-cGAMP and *Ps*Cap5–3′2′-c-diAMP complexes (Fig. 4a,b). These contacts are extensive, and their geometry is such that it will not easily accommodate deviations from the 3′2′ linkage and provides a basis for understanding why ligands closely related to 3′2′-cGAMP such as 3′3′-cGAMP and 2′3′-cGAMP are unable to activate *Ps*Cap5. Substitution by alanine of the three residues (His138, Arg242, Ser277) that directly contact the phosphate groups of the ligand completely eliminates the DNA cleavage activity of *Ps*Cap5 (Extended Data Fig. 5a,b).

Structure of *Ps*Cap5 in the absence of a ligand

We succeeded in crystallizing full-length *Ps*Cap5 in the absence of an activating ligand. The apo crystal contains *Ps*Cap5 as a dimer. This is consistent with our mass photometry data showing that *Ps*Cap5 exists as a dimer in the absence of an activating ligand and that it transitions to a tetramer only in the presence of an activating ligand. The apo dimer is arranged in a crisscross manner with protomers A and B related by twofold symmetry and assume almost identical conformations (r.m.s.d. of 1.57 Å for 354 Cαs) (Fig. 5, top panel). The SAVED domains are relatively far apart with most of the contacts between them limited to loop residues 201–208 (Extended Data Fig. 7a). We refer to the SAVED domains in this apo state as in an 'open' configuration, as opposed to the closed configuration with a ligand. In the structure of apo *Ll*Cap5, the SAVED domains are splayed even further apart without any contacts between them, whereas in the apo *As*Cap5 structure they are apposed close

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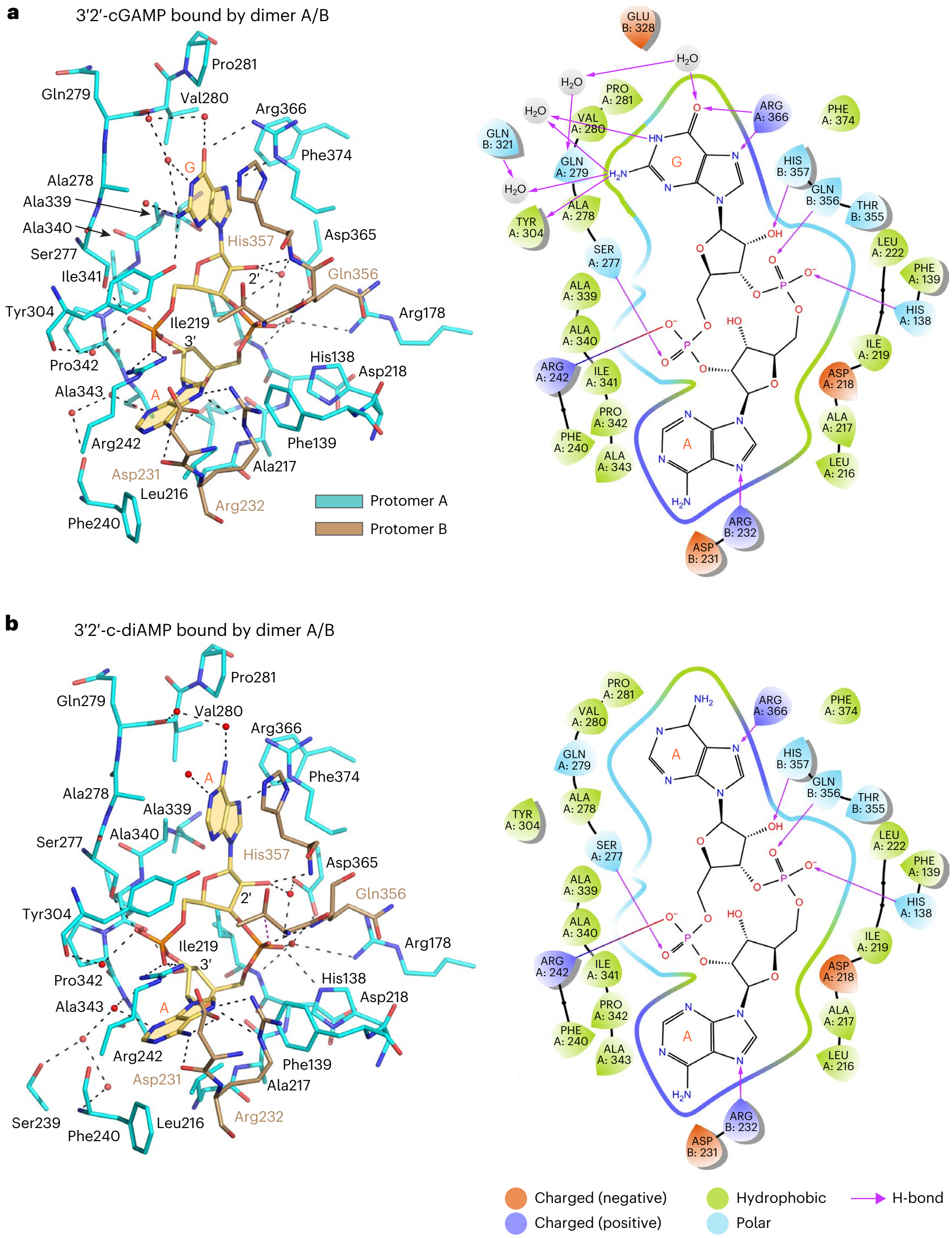

Fig. 4 | Recognition of the 3′2′-cGAMP and 3′2′-c-diAMP ligands by *Ps*Cap5 and interaction diagrams. a,**b**, Interactions of protomers A and B with 3′2′-cGAMP (**a**) and 3′2′-c-diAMP ligands (**b**). The ligands, shown as sticks with carbon atoms colored yellow, are bound between the SAVED domains of protomers A and B (and C and D) in the *Ps*Cap5-activated tetramer. The protein residues involved in ligand recognition are shown as sticks with the carbon atoms of the residues colored in cyan for protomer A and in beige for protomer B. The SAVED of protomer A (and C) comprises most of the ligand binding pocket with the SAVED of protomer B (and D) using a different interface to form a smaller lid to fully enclose the ligand. For the recognition of 3′2′-cGAMP, SAVED of *Ps*Cap5 protomers A and B make hydrophobic and polar contacts that together provide specificity for the G and A nucleobases and the asymmetric sugar-phosphate linkages between them. Interactions from SAVED of protomer A involve predominantly residues that stem from loops that precede several α-helices (α5, α8, α9 and α11) and β-strands (β9) that coalesce at the interface between the two CARF modules. There is an interesting division of labor, wherein residues from CARF1 (His138, Phe139, Leu216, Ala217, Ile219, Phe240 and Arg242) cluster around the A nucleobase and residues from CARF2 (Ala278, Val280, Pro281 and Phe374) surround the G nucleobase. However, it is primarily residues from CARF2 (Ser277, Ala339, Ala340, Ile341, Pro342, Ala343, Asp365) that encompass the sugar-phosphate linkages between the nucleobases. Residues from SAVED of protomer B (Arg232, Asp231, Gln356 and His357) add to the interactions with 3′2′-cGAMP; for example, His357 stacking against the A nucleobase.

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together in an asymmetric 'ajar' conformation[10]. The *Ps*Cap5 structure further highlights the mobility of the SAVED domains in Cap5 in the absence of a ligand. The interfaces between the HNH domains and the connector helices in our structure are different from the ligand-bound state (Extended Data Figs. 4a,b and 7a–c). The HNH domains lack a catalytic Mg^{2+} and the general base His56 is disordered (Extended Data Fig. 8). Together, these changes underscore the extraordinary remodeling of *Ps*Cap5 that takes place on ligand binding (Fig. 5).

Transition of *Ps*Cap5 from apo to ligand-bound state

On ligand binding the SAVED domains close and rotate relative to the HNH endonuclease domains and the connector helices. Compared to the apo state, the SAVED domain of protomer A rotates by 160° and presents its opposite surface to the SAVED domain of protomer B. For example, whereas the β9-α10 structural element (selected for visual reference) in the apo protomer A faces outward, it faces inward in the ligand-bound state (Fig. 5, top and middle panels, and Extended Data Fig. 9a,b). Protomer B on the other hand rotates by lesser 82° with the β9-α10 structural element facing outward (Fig. 5, top and middle panels, and Extended Data Fig. 9a,c). This unequal or asymmetric rotation of the two SAVED domains (relative to the apo state) augments the recognition of asymmetrical ligands by presenting two different surfaces of the SAVED domains. Approximately 1,540 $Å^2$ of surface area is buried between the two SAVED domains, compared to only 600 $Å^2$ in the apo state. The SAVED domain itself undergoes only minor local adjustments on engaging one or the other surfaces for ligand binding (Extended Data Fig. 2b). The two ligand-bound crisscross dimers in turn expose a surface that lends to formation of the active tetramer with the HNH domains of protomers A and C activated through secondary structure remodeling and juxtaposed for DNA cleavage (Fig. 5, bottom panel). By contrast, the HNH domains of protomers B and D are positioned on the outside of the tetramer assembly in catalytically inactive states, which resemble the structure of HNH domains in the apo state in lacking a catalytic Mg^{2+} (Extended Data Fig. 8), among other changes. Overall, the tetramer is held together by extensive protein–protein interactions, wherein each protomer makes contacts with the all three of the other protomers (Fig. 2a and Extended Data Figs. 4a,b and 7b,c).

Extrinsic activation of the CBASS limits bacterial growth

The CBASS is common in environmental and pathogenic bacteria. An outstanding question is whether the cell killing activity of CBASS effectors such as Cap5 can be leveraged or harnessed to limit infection by pathogenic bacteria. We provide here proof-in-principle that the CBASS can be extrinsically activated to limit bacterial growth and infection. Briefly, we incubated *Escherichia coli* cultures carrying a plasmid encoding wt*Ps*Cap5 (wt, wild-type) with and without 3′2′-cGAMP in the culture medium and then serially diluted and spot-plated the cultures. The bacterial growth of wt*Ps*Cap5-expressing culture is severely restricted by 3′2′-cGAMP and no bacterial growth is evident in the 10^{-1}–10^{-5} dilution range (Fig. 6a, top left panel). This occurs even in the absence of isopropyl-β-D-1-thiogalactopyranoside (IPTG) due to leaky expression of the plasmid (Fig. 6b). As a control, when we expressed a *Ps*Cap5 H91A/D95A mutant inactive for the HNH endonuclease activity (Extended Data Fig. 5b), growth of the bacterial culture was unaffected (Fig. 6a, top right panel). Thus, under our experimental conditions, high levels of wt*Ps*Cap5 protein are produced with or without IPTG induction (Fig. 6b) and the extrinsic addition of 3′2′-cGAMP limits the bacterial growth. As a further control, the addition of 3′3′-cGAMP and 2′3′-cGAMP, which do not activate *Ps*Cap5 for DNA digestion, also did not restrict the growth of the wt*Ps*Cap5-expressing culture (Fig. 6a, lower panels). The cell growth of 3′2′-cGAMP/wt*Ps*Cap5 cultures recovers with a longer 16 hour incubation time (Extended Data Fig. 10a). DNA sequencing of plasmids isolated from the single bacterial colonies revealed loss of either the entire wt*Ps*Cap5-coding sequence or of the HNH endonuclease domain (Extended Data Fig. 10b,c).

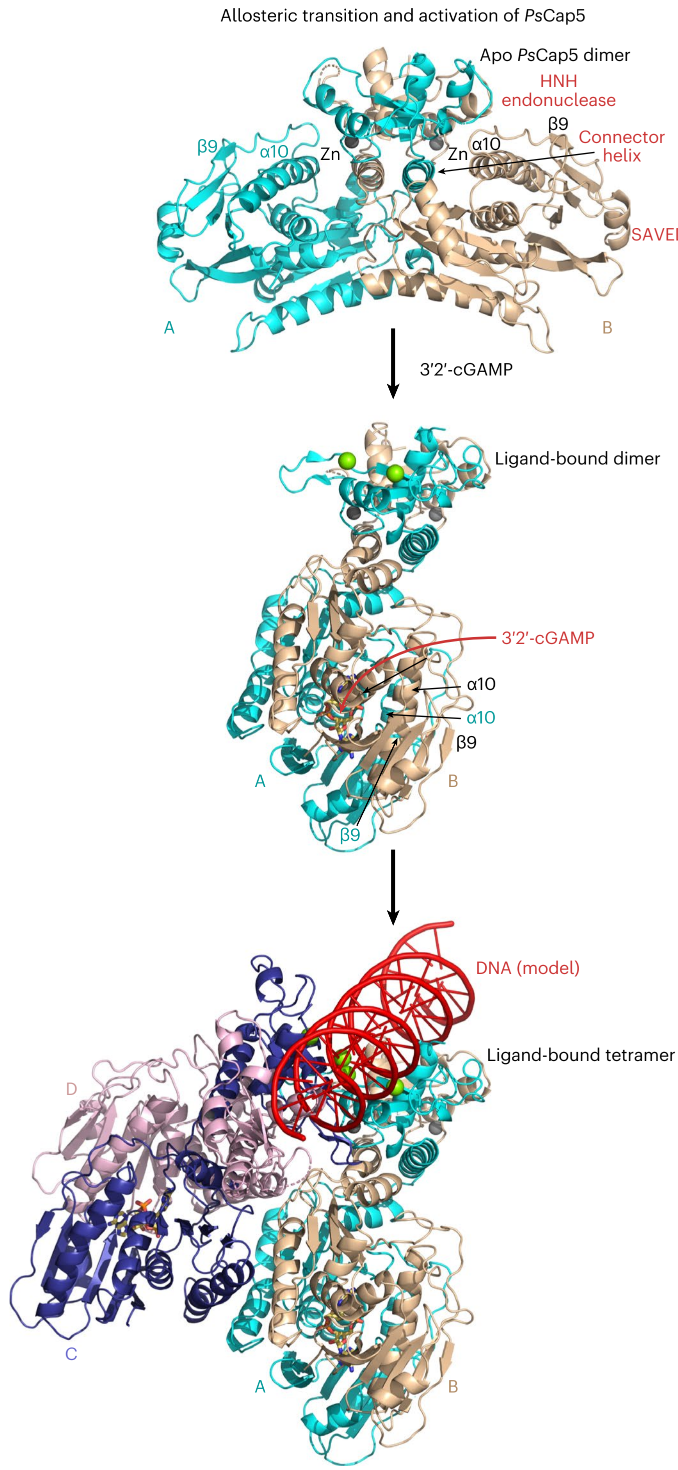

Fig. 5 | Structure of apo *Ps*Cap5 dimer and transition to the ligand-bound activated tetramer. The top panel shows the structure of apo *Ps*Cap5 crisscross dimer with the SAVED domains in the open conformation. The middle panel shows the binding of the activating ligand triggers closure of the SAVED domains by rotation relative to the HNH endonuclease domains and the connector helices. The SAVED of protomer A rotates by 160° and presents the opposite surface to the SAVED of protomer B. Protomer B on the other hand rotates by 82° with the β9-α10 structural element facing outward. The bottom panel shows the two ligand-bound dimers form the tetramer (A/B and C/D) with the HNH endonuclease domains of protomers A and C juxtaposed and activated for DNA cleavage. The structures of the apo dimer (protomers A and B), ligand-bound dimer (protomers A and B; one of the two dimers in the tetramer) and the dimer-of-dimers tetramer (protomers A/B and C/D) are shown in the same view with the apo and tetramer structures superimposed by the HNH endonuclease domains of protomers A and B.

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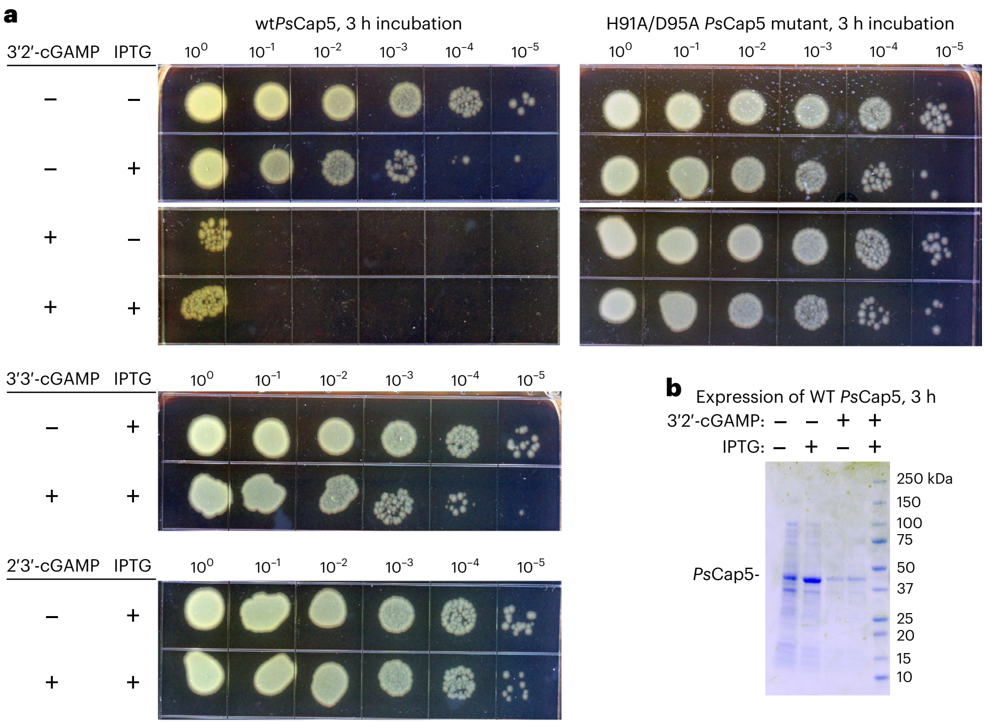

Fig. 6 | Extrinsically supplied 3′2′-cGAMP limits growth of *Ps*Cap5-expressing bacteria. a, Serial dilution of *E. coli* culture carrying plasmid encoding wt*Ps*Cap5 or the catalytically inactive H91A/D95A mutant in the absence or presence of IPTG and *Ps*Cap5-activating 3′2′-cGAMP ligand or nonactivating 3′3′-cGAMP and 2′3′-cGAMP cyclic dinucleotides. After reaching an OD_{600} of roughly 0.6, the cultures were incubated for 3 h with or without IPTG and dinucleotides as indicated, serially diluted and spot-plated. Pronounced reduction in bacterial growth is observed with 3′2′-cGAMP and the wt*Ps*Cap5 plasmid. The plates are representatives of at least three biologically independent experiments. **b**, High levels of expression of wt*Ps*Cap5 protein are observed either with or without induction with IPTG. The gel is a representative of at least three independent experiments.

Discussion

We present here high-resolution structures of a full-length Cap5 from *P. syringae* in complex with second messengers. We show here that the key to *Ps*Cap5 activation is a dimer to tetramer transition. The binding of a second messenger to *Ps*Cap5 dimer triggers an open to closed transformation of the SAVED domains within each dimer and, in turn, furnishes a surface for the assembly of the *Ps*Cap5 tetramer (dimer-of-dimers). This configurational change in the SAVED domains propagates to the HNH endonuclease domains, converting two of them into catalytically competent states and bringing them in close proximity for DNA degradation. Change in oligomerization is emerging as a common theme in CBASS effectors, although the nature of the oligomer varies. From electron microscopy analysis, the CBASS effectors Cap4 (restriction endonuclease (RE)-SAVED fusion)[9], Cap12 (NAD^+ cleaving TIR-STING fusion)[21] and TIR-SAVED[12] form filaments in the presence of a second messenger, and from crystallographic analysis the CBASS effector NucC (RE-like fold) transitions from a trimer to a hexamer[8]. NucC lacks a domain (such as SAVED or STING) dedicated for nucleotide binding. The second messenger binds instead at the periphery of the NucC RE-fold and the conformational changes induced are much subtler than the gross domain movements observed for *Ps*Cap5. Overall, a dimer to tetramer transition observed with *Ps*Cap5 highlights the diversity of multimerization mechanisms for activating CBASS effectors. This appears to be a common mechanism for Cap5 effectors since the related *As*Cap5 and *Ll*Cap5 proteins also do not produce higher-order complexes on ligand binding as shown by gel filtration[10]. The closed symmetry of *Ps*Cap5 tetramer is counter to the notion that CBASS (and other innate immune) effectors containing a SAVED domain are activated by open symmetry multimerization leading to filaments[9,10,12].

We show here that the primary binding site for the second messenger is at the interface between the two SAVED domains within each *Ps*Cap5 dimer of the activated tetramer. The second messenger acts effectively as a glue to close the SAVED domains within each dimer for subsequent tetramerization. We find that one SAVED domain makes most interactions with the second messenger while the other forms a lid over it to enclose it. This asymmetry in interactions occurs because the two SAVED domains undergo very different rotations (160° and 82°) compared to the unbound structure, and thereby present different surfaces for interactions with the second messenger. The ajar conformation of SAVED domains observed for apo *As*Cap5 (ref. 10) could represent an intermediate dynamic state of *Ps*Cap5 for ligand sampling but this remains to be tested. As first shown for Cap4 (ref. 9), the SAVED domain is a fusion of two pseudo-symmetrically related CARF modules that obviates the need for twofold symmetric second messengers. The fact that *Ps*Cap5 uses two SAVED domains (or four CARF modules) to recognize a cyclic dinucleotide expands the repertoire of second messenger signals that can be recognized by Cap5 and other CBASS effectors. Also, because the two SAVED domains undergo drastically different rotations on nucleotide binding, this further averts the requirement for internally twofold symmetric second messengers. This is important as some of the key signaling molecules for normal cellular processes in bacteria are twofold symmetric cyclic dinucleotides such as 3′3′-c-diGMP and 3′3′-c-diAMP. We show that neither of these symmetric cyclic dinucleotides can activate *Ps*Cap5, guarding the effector from inadvertent activation by signals generated during normal cellular processes.

P. syringae is a plant pathogen. The CBASS is also common in bacteria that infect humans. We provide here proof-in-principle evidence that the cell killing activity of CBASS effectors such as Cap5 can be harnessed to limit infection by pathogenic bacteria. We show that extrinsic addition of the activating ligand 3′2′-cGAMP to an *E. coli* culture expressing *Ps*Cap5 severely restricts cell growth, suggesting that the cyclic nucleotide can cross even the double membrane cell wall of a Gram-negative bacterium such as *E. coli* to activate Cap5. In all, the identity of CBASS effectors and their activating cyclic nucleotides may offer new therapeutic opportunities in controlling infections by pathogenic bacteria.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41594-024-01220-x.

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Methods

Protein expression and purification

The DNA sequence coding for full-length (residues 1–388) *P. syringae* (*Ps*) Cap5 (TRN83959.1) was codon optimized for protein expression in *E. coli* and then synthesized and inserted between BamHI and HindIII cloning sites of a modified pET28b(+) vector (Novagen) to produce N-terminally tagged His6-SUMO (small ubiquitin-like modifier) fusion protein (GenScript). The plasmid was transformed into *E. coli* BL21 (DE3) competent cells (Agilent). The cells were grown in Luria-Bertani (LB) medium with 30 μg ml^{-1} kanamycin at 37 °C to an optical density (OD_{600}) of 0.6, the temperature was then decreased to 18 °C and protein expression induced by the addition of 0.4 mM IPTG. After overnight growth, the bacterial cells were pelleted by centrifugation, resuspended in 25 mM TRIS-HCl pH 8.0, 20 mM imidazole, 500 mM NaCl, 10% glycerol, 2 mM β-mercaptoethanol (BME) with a cOmplete EDTA-free protease inhibitor cocktail tablet (Roche) and lysed by sonication. The lysate was centrifugated at 38,724*g* for 1 h and the protein was purified from the soluble fraction by a nickel chelating His60 Ni Superflow 5 ml column (Takara Bio Inc.) connected to an ÄKTA FPLC system (GE Healthcare Life Sciences). The protein was eluted from the column with 25 mM TRIS-HCl pH 8.0, 300 mM imidazole, 300 mM NaCl, 10% glycerol, 2 mM BME buffer. The His6-SUMO tag was cleaved with Ulp1 protease during an overnight dialysis into 25 mM TRIS-HCl pH 8.0, 300 mM NaCl, 2 mM BME buffer. The protein was passed through the His60 Ni column to bind the cleaved His6-SUMO tag and His6-tagged Ulp1. *Ps*Cap5 protein was then purified by gel filtration on a Superdex 75 column (GE Healthcare) and concentrated in 25 mM TRIS-HCl pH 8.0, 250 mM NaCl and 2 mM tris(2-carboxyethyl)-phosphate (TCEP) to ~32 mg ml^{-1}, and stored in aliquots at −80 °C. The site-directed mutagenesis of the *Ps*Cap5 expression plasmid was conducted by Genscript and His56Ala, His91Ala-Asp95Ala, Glu53Ala-Asp92Ala, Asn185Trp-Asp188Arg, Asn185Arg-Asp188Arg and His138Ala-Arg242Ala-Ser277Ala *Ps*Cap5 mutant proteins were expressed and purified as described above.

DNA plasmid digestion assays

For DNA cleavage assays we used a pET-29b(+) plasmid with human *FEN1* gene residues 2–366 inserted between NdeI and XhoI sites for a total of 6.4 kbp (kilobase pairs). *Ps*Cap5 protein was incubated with (or without) the cyclic nucleotide ligands for 15 min on ice in 10 mM Tris-HCl pH 7.5, 25 mM NaCl, 10 mM $MgCl_2$ and 1 mM TCEP. The reactions were initiated by addition of 500 ng of the DNA plasmid. The total reaction volume was 25 μl. For the identification of putative activating ligands, we used 500 nM protein and 500 nM ligand concentrations; 3′–5′/3′–5′ (3′3′) linked cyclic di-, tri- and tetra-nucleotides as well as mixed-linked 2′–5′/3′–5′ (2′3′) and 2′–5′/2′–5′ (2′2′) cyclic purine dinucleotide ligands were obtained from the BIOLOG Life Science Institute. For the evaluation of the activating ligand concentration the protein concentration was reduced to 50 nM and the ligand concentrations was varied in the 10^{-2}–10^{4} nM range as indicated on the corresponding figures. The reactions were incubated for 20 min at 37 °C and stopped by addition of 6× loading buffer (60 mM EDTA pH 8.0, 20 mM Tris-HCl pH 7.5, 30% glycerol, 0.1% SDS, 0.012% xylene cyanol). The DNA cleavage products were separated by electrophoresis on 1.5% agarose gel with TAE buffer, and the 1 kbp DNA ladder from New England Biolabs was used as a DNA mobility weight marker. After running the gel for 30 min at 100 V, the gels were stained with 10 μg ml^{-1} ethidium bromide in water for 30 min and then destained in water for 10 min. The gels were photographed with a GelDoc-It2 imaging system running VisionWorks v.8.17.16133.9147 software. The black and white colors of the original gels were inverted for figure clarity. The unprocessed images of the gels are provided in the Source data.

Mass photometry

Mass photometry experiments were performed on a Refeyn OneMP mass photometer (Refeyn Ltd). The datasets were recorded with the Acquire^{MP} v.2023 R1.1 and analyzed with the Discover^{MP} v.2023 R1.2 software. To form a sample chamber, self-adhesive silicon gasket (4–6 wells, ~15 μl volume each) was placed on a cleaned and dried borosilicate glass microscope cover slide. To conduct a measurement, the glass slide and silicone gasket assembly was placed on the instrument's objective and centered on a single well, then 12 μl of sample buffer was added to the well and the focal position of the glass surface was determined and held constant using an autofocus system. Next, 1 μl of protein sample was added to the well (13-fold dilution) and mixed with the buffer. A 90 s long video was recorded immediately after the dilution. A fresh well was used for each measurement and the measurement was repeated at least three times for each sample. An aliquot of *Ps*Cap5 was defrozen on ice and serially diluted in the sample buffer (25 mM Tris-HCl pH 8.0, 250 mM NaCl, 2 mM TCEP and 5 mM $MgCl_2$) so that the number of detected events (particle counts) during a measurement was roughly 3,000 to 5,000 per 90 s long video for an optimum data acquisition and processing. A contrast-to-mass linear calibration curve was created using BSA (monomer and dimer) and apoferritin protein standards diluted in the sample buffer. To calculate the molecular weight of the main species observed on the particle counts versus molecular mass distribution histograms we used the Gaussian function in the Discover^{MP} software. The final concentration of *Ps*Cap5 in a sample well was 5.8 nM and the ligand concentration (if present) was 25 μM. We noticed that the fraction of the tetramer peak in the presence of the activating ligands 3′2′-cGAMP or 3′2′-c-diAMP increased if the protein was incubated with the ligand in a test tube before the mass photometry measurement. The fraction of tetramers reached the maximum at ~15 min after the sample mixing. Further incubation (up to 3 h, on ice or at room temperature) did not increase the proportion of the tetramers. Decreasing NaCl salt concentration in the buffer to 100 mM did not affect mass distribution.

Crystallization, data collection and structure determination

Initial crystallization conditions for apo *Ps*Cap5 were determined with matrix screens (Hampton Research, Molecular Dimensions and Qiagen) set up with OryxNano crystallization robot (Douglas Instruments) using 10 mg ml^{-1} protein in buffer containing 25 mM Tris-HCl pH 8.0, 250 mM NaCl, 2 mM TCEP and 5 mM $MgCl_2$ at 20 °C. Small crystals appeared after 3 days of incubation in 1 M trisodium citrate with 0.1 M imidazole pH 8.0. Large rectangular bipyramid crystals up to ~0.4 × 0.2 × 0.2 mm in size were optimized to grow in 1 M trisodium citrate with 0.1 M TRIS-HCl pH 8.0 in hanging drop vapor diffusion experiments; large triangular prism shaped crystals were also observed in different drops under the same conditions. The crystals cryoprotected in mother liquor–glycerol mixture (1.26 M trisodium citrate, 0.1 M TRIS-HCl pH 8.0 and 10% glycerol) diffracted poorly (~12–15 Å resolution) with synchrotron radiation under cryogenic conditions (NSLS-II 17-ID-1 and 17-ID-2 beamlines at the Brookhaven National Laboratory and the NE-CAT 24-ID-E beamline at the Argonne National Laboratory). To improve the diffraction, we screened various cryoprotective mixtures. The use of 1.4 M trisodium citrate with 0.375 M sodium formate and 50 mM TRIS-HCl pH 8.0 as a cryoprotectant resulted in consistent resolution improvement to ~5–7 Å, and ~2 h of soaking in the cryoprotectant further improved the resolution to ~4–6 Å. About 325 crystals were screened in total and the best crystal diffracted to 3.16 Å at the 24-ID-E beamline. This apo *Ps*Cap5 crystal belonged to the $P4_12_12$ space group with $a = b = 159.4$ Å, $c = 433.9$ Å $\alpha = \beta = \gamma = 90.0°$ unit cell. We derived relaxed AlphaFold2 (ref. 23) models using the AlphaFold2 advanced beta version for *Ps*Cap5 monomer and dimer using the ColabFold server[24] for molecular replacement by the Phaser[25] module in Phenix[26] software suite v.1.20.1-4487. The models were processed with phenix.process_predicted_model before molecular replacement search. The best search results were obtained by using a SAVED domain and a predicted HNH endonuclease domain dimer (with some regions manually trimmed to eliminate clashes)

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as the search models. Phaser was able to successfully place six copies of the HNH endonuclease domain dimers and ten copies of the SAVED. We docked the remaining two copies of SAVED into the density modified map generated in Phenix for a total of 12 copies of *Ps*Cap5 protein (six dimers) in the asymmetric unit. For refinement in phenix.refine, we used a *Ps*Cap5 monomer from a high-resolution structure with 3′2′-c-diAMP (below) obtained at that time as a reference model. We were able to refine the apo *Ps*Cap5 structure to R_{work}/R_{free} of ~24.0/29.5% but it had a pronounced number of geometry outliers likely due to differences in conformations in many regions of the apo structure compared to the ligand-bound tetramer. In addition, several regions of the structure had less defined electron density. To improve the fit and geometry of the apo *Ps*Cap5 model, we remodeled each of the six apo dimers by removing regions with poorly defined electron density and using the resultant model as a template for an AlphaFold-multimer[27] ColabFold v.1.5.3 prediction with the ColabFold server, as recently described[28]. Rebuilt dimers were placed back into the model by superimposing onto their template counterparts with a least squares fit of main chain atoms followed by with rigid body and atomic displacement parameters with phenix.refine. This increased the fraction of residues in the Ramachandran favored regions from ~88 to 96%. The rebuilt model was then relaxed into the sharpened experimental electron density map with the molecular dynamics refinement tool ISOLDE[29]. The refinement was completed with Phenix-refine and the final model was refined to 3.16 Å resolution with R_{work} and R_{free} values of 25.0 and 29.9%, respectively, and displayed good stereochemistry with 95% of residues in the Ramachandran favored regions (Table 1).

To obtain crystals of activated *Ps*Cap5 we rescreened *Ps*Cap5 in the presence of 1 mM 3′2′-cGAMP and 3′2′-c-diAMP cyclic dinucleotides with the matrix screens. The 3′2′-cGAMP-liganded crystals were optimized to grow as thick large rectangular plates in 0.1 M Bis-Tris pH 5.5–6.0 with 0.2 M $MgCl_2$ and PEG3350 19–20% with several rounds of seeding. The 3′2′-c-diAMP-liganded crystals grew under similar conditions as hexagonal plates. The crystals were cryoprotected in mother liquor solution with 24% PEG3350 and stepwise increase of glycerol content to 20%. Most of these crystals diffracted to high resolution at NSLS-II 17-ID-1 and 17-ID-2 beamlines at the Brookhaven National Laboratory. The 3′2′-cGAMP-liganded structure was refined to 1.87 Å resolution and has two tetrameric assemblies in the asymmetric unit (space group $P2_1$ with a = 67.97 Å, b = 295.68 Å, c = 84.05 Å, $\alpha = \gamma$ = 90.0° and β = 113.8° unit cell). The 3′2′-c-diAMP liganded structure crystallized in two space groups: C2 with a = 145.19 Å, b = 80.52 Å, c = 406.84 Å, $\alpha = \gamma$ = 90.0° and β = 90.6° unit cell; and $P3_22$ with the $a = b$ = 83.54 Å, c = 401.96 Å, $\alpha = \beta$ = 90.0° and γ = 120° unit cell. The C2 crystal form diffracted to 1.79 Å and has three tetramers in the asymmetric unit, whereas the $P3_22$ crystal form diffracted to 2.11 Å and has a single tetramer in the asymmetric unit. We first solved the 3′2′-c-diAMP liganded structure with a single tetramer complex in the asymmetric unit by molecular replacement using the AlphaFold2 models as described above for the apo structure and then used the refined model to solve the other structures. The structures were refined in Phenix-refine with manual rebuilding in Coot[30]. The 3′2′-c-diAMP-liganded C2 crystal form has been refined to 1.79 Å with R_{work}/R_{free} of 17.9/22.5%, and the $P3_22$ to 2.11 Å with R_{work}/R_{free} of 19.0/23.7%. The 3′2′-cGAMP-liganded structure at 1.87 Å resolution has R_{work} and R_{free} values of 21.0 and 25.3%, respectively. The data collection and refinement statistics are summarized in Table 1 for all the structures. X-ray diffraction data were collected using LSDC v.2.0.2 computer software available at the 17-ID-1 and 17-ID-2 beamlines at NSLS-II at the Brookhaven National Laboratory and a custom browser-based general user interface at the NE-CAT 24-ID-E beamline at the Advanced Photon Source, Argonne National Laboratory. All crystallographic datasets were processed with autoPROC[31] v.1.0.5. All molecular graphic figures were prepared using PyMOL v.2.5.5. (Schrödinger LLC). Surface areas were calculated using PDBePISA v.1.52 (the protein interfaces, surfaces and assemblies service at the European Bioinformatics Institute; http://www.ebi.ac.uk/pdbe/prot_int/pistart.html)[32]. For secondary structure analysis and comparison, we used the DSSP algorithm[33] at https://2struccompare.cryst.bbk.ac.uk/index.php (ref. 34). The interaction diagrams for the protein contacts with the ligands were produced with the Maestro module v.11.8.012 in the Schrodinger suite of programs (Schrodinger LLC) with some manual edits. Protein topology diagrams generated in Pro-origami (https://www.ibg.kit.edu/protein_origami/) were used as a guide to produce the diagrams presented in the figures.

Modeling of DNA in the active site of *Ps*Cap5 tetramer

A model of HNH endonuclease domains A/C interactions with DNA is based on the structure of Colicin E9 HNH endonuclease monomer with DNA (Protein Data Bank (PDB) ID 1V15; ref. 22). The ββα module of Colicin E9 HNH endonuclease monomer with a scissile DNA strand was superimposed to the ββα modules of the HNH endonucleases of protomers A and C, as shown in Fig. 3f. This produced a 5 base pair (bp) long DNA duplex with a slightly wider minor groove than in ideal B-DNA. To generate a longer DNA model (Fig. 5 and Extended Data Fig. 6), a 23 bp ideal B-DNA generated in Coot was superimposed to the 5 bp DNA duplex, the 5 bp fragment was then inserted into the ideal B-DNA.

Growth and analysis of *Ps*Cap5-expressing bacteria in the presence of cyclic dinucleotides

The *Ps*Cap5-coding DNA sequence optimized for protein expression in *E. coli* (Genscript) was inserted between the NdeI and HindIII cloning sites of the pET28b(+) vector. The plasmid was transformed into *E. coli* BL21 (DE3) competent cells. Transformants were grown in LB medium with 50 µg ml^{-1} kanamycin at 37 °C to an OD_{600} of ~0.6. Then, 1 ml of the culture was transferred to a 5 ml culture tube and incubated at 37 °C for 3 h with or without 0.1 µM IPTG and/or 0.1 mM cyclic dinucleotides as indicated in Fig. 6. The bacterial cultures were then serially diluted from 10^0 to 10^{-5} and 5 µl of each spotted onto LB agar petri plates supplemented with kanamycin. After overnight incubation, the plates were scanned with an Epson Reflection V600 scanner. Control experiments were conducted with bacteria carrying the catalytically inactive H91A/D95A *Ps*Cap5 mutant plasmid (Genscript). Growth of the remaining 3 h-incubated cells was continued for 16 h in total and spotted on plates as described above. Plasmid DNA from the 16 h cultures was isolated and transformed into XL10-Gold *E. coli* strain (Agilent) with selection for kanamycin resistance. Next, 14 single bacterial colonies from 3′2′-cGAMP/IPTG exposed cultures with wt*Ps*Cap5 plasmids were cultured separately and the plasmid DNAs were then isolated by using QIAprep miniprep kit (Qiagen). The analysis by agarose gel electrophoresis indicated smaller sizes for all recovered plasmids compare to the original wt*Ps*Cap5-coding plasmid. To evaluate the lost DNA regions, the plasmids were digested with a set of restriction enzymes and based on the observed fragments DNA primers were designed for Sanger DNA sequencing (Azenta). DNA sequencing revealed nine unique deletion plasmids, which are listed in Extended Data Fig. 10c. Sequencing of two DNA plasmids isolated from the wt*Ps*Cap5-expressing cultures incubated for 16 h with IPTG in the absence of 3′2′-cGAMP indicated no mutations affecting the *Ps*Cap5 protein.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Associated data are provided in Extended Data Figs. 1–10. The unprocessed images of the electrophoretic gels are supplied in the source data. Atomic coordinates and structure factors for *Ps*Cap5 with 3′2′-cGAMP, 3′2′-c-diAMP (with three and one tetramer assemblies in the asymmetric unit) and without a ligand are available in the PDB under accession codes 8FMH, 8FMG, 8FMF and 8FM1, respectively. Source data are provided with this paper.

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Author contributions

O.R. and A.K.A. designed the experiments. O.R. performed mass photometry, crystallization, structure solution and refinement for all the structures. D.S. performed DNA plasmid digestion assays and growth of *Ps*Cap5-expressing bacteria in the presence of cyclic dinucleotides. D.F.K. contributed to molecular replacement phasing of the apo *Ps*Cap5 structure and performed advanced refinement for the structure (AlfaFold2 and ISOLDE) and assisted with refinement for the other structures. J.K. assisted in X-ray data collection and mass photometry experiments. A.B. conducted wild-type protein expression and purification, D.S. expressed the mutant *Ps*Cap5 proteins and O.R. purified these proteins. A.K.A. guided the project. O.R. and A.K.A. wrote the paper.

Competing interests

The authors declare no competing interests.

Additional information

Extended data is available for this paper at https://doi.org/10.1038/s41594-024-01220-x.

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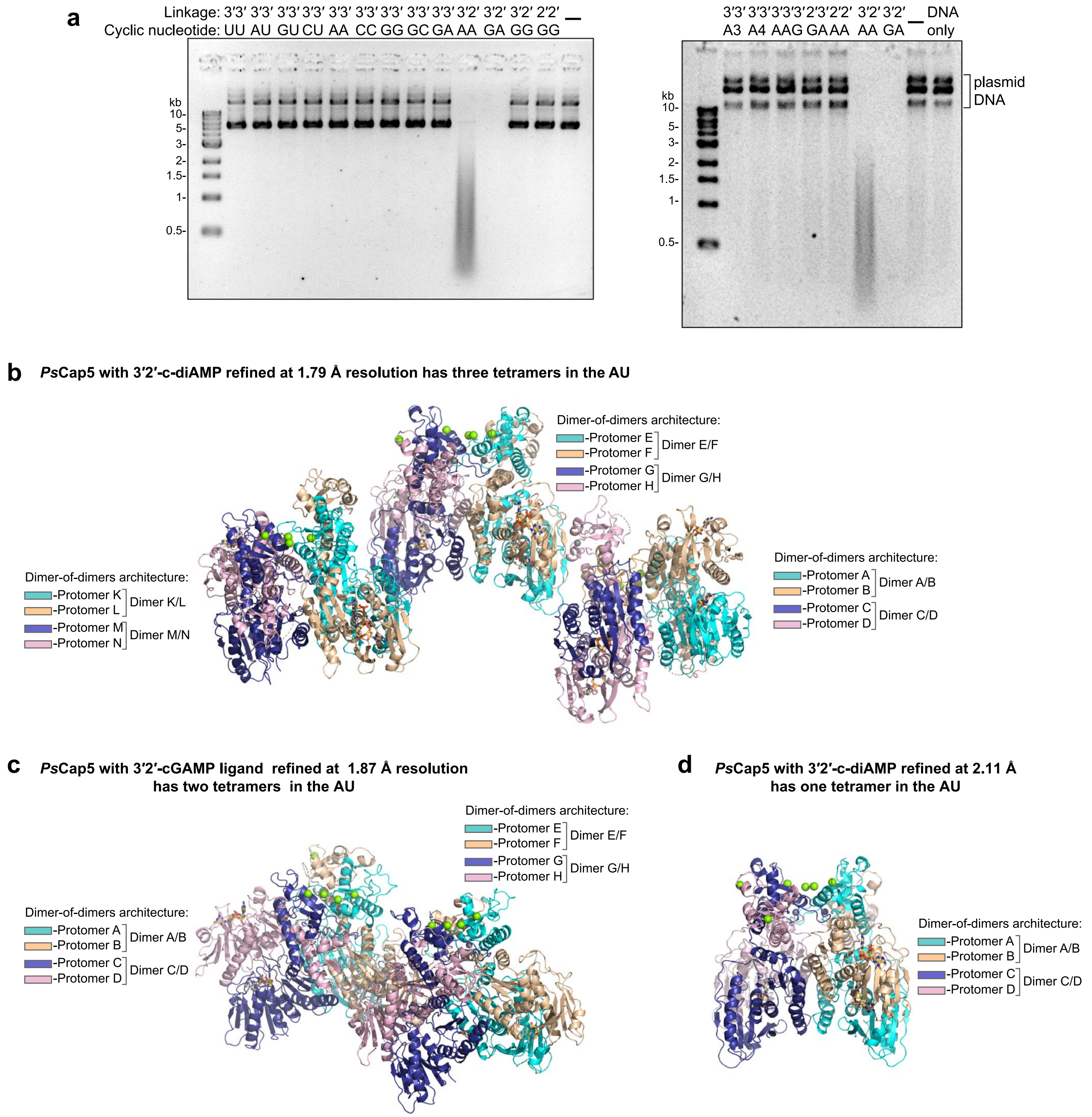
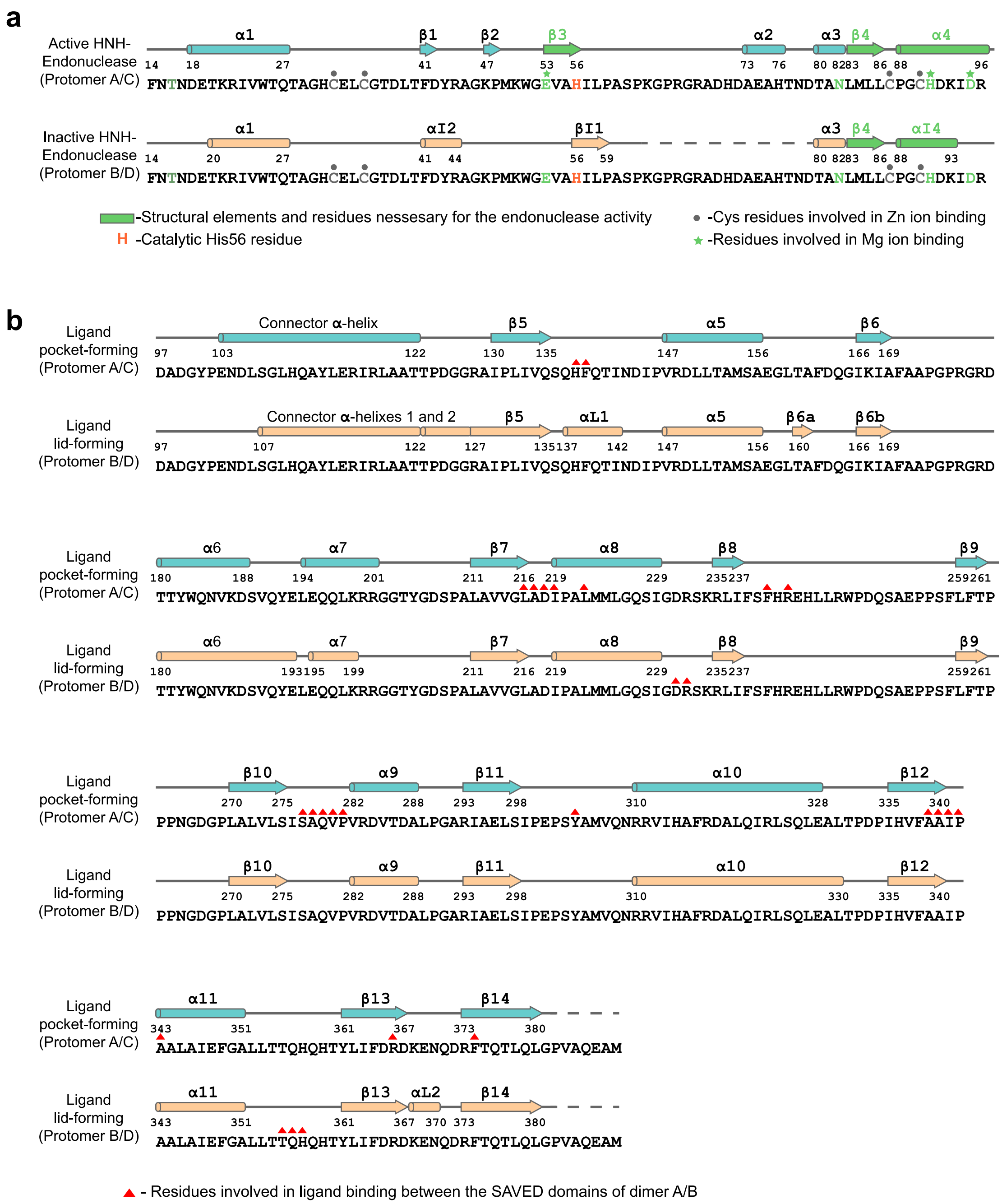


Extended Data Fig. 1 | Plasmid DNA digestion by *Ps*Cap5 in the presence of cyclic nucleotides and structures of the activated ligand-bound *Ps*Cap5 tetramers. **a**, Plasmid DNA digestion assay with di-, tri- and tetra- cyclic nucleotides with 3′3′, 2′3′ and 2′2′ linkages. Related to Fig. 1b that shows DNA digestion with only purine dinucleotide ligands. Plasmid digestion is observed only with 3′2′-cGAMP (3′2′GA) and 3′2′-c-diAMP (3′2′AA). Concentrations of *Ps*Cap5 and the ligands are 500 nM. The DNA cleavage products are analyzed by agarose gel electrophoresis followed by staining with ethidium bromide. The black and white colors are inverted for clarity. Plasmid DNA includes closed and open circle species. Each gel is a representative of at least three independent experiments. **b**, The 3′2′-c-diAMP liganded structure in C2 space group with a = 145.19 Å, b = 80.52 Å, c = 406.84 Å, α = γ = 90.0° and β = 90.6° unit cell is refined to 1.79 Å resolution and has three tetramer assemblies in the asymmetric unit (AU). **c**, The 3′2′-cGAMP containing structure in $P2_1$ space group and with a = 67.97 Å, b = 295.68 Å, c = 84.05 Å, α = γ = 90.0° and β = 113.8° unit cell is refined to 1.87 Å resolution and has two tetramer complexes in the AU. **d**, The 3′2′-c-diAMP liganded structure in $P3_22$ space group with the a = b = 83.54 Å, c = 401.96 Å, α = β = 90.0° and γ = 120° unit cell is refined to 2.11 Å and has a single tetramer assembly in the AU.



Extended Data Fig. 2 | Sequence and alternate secondary structures of protomers A (and C) and B (and D) in ligand-bound *Ps*Cap5 tetramer. **a**, Catalytically active and inactive conformations of the HNH endonuclease domains in the tetramer. HNH endonucleases of protomers A and C have the active conformation with the characteristic ββα-topology of HNH endonucleases colored in green. Catalytic His56 is highlighted in red. HNH endonucleases of protomers B and D have inactive conformation and the ββα-topology is disrupted. Residues that are involved in coordination of the catalytic and structural Mg^{2+} are indicated by green stars. Cys32, Cys35, Cys87 and Cys90 coordinate a structural Zn^{2+} ion and are indicated by the gray dot. Residues 63-78 of the inactive HNH endonuclease (protomers B and D) are disordered. The N-terminal residues 1-13 are disordered in all protomers of the tetramer. **b**, The SAVED domains of dimer A/B (and C/D) in the tetramer. The SAVED domain of protomer A (and C) forms most of the ligand binding pocket (secondary structure elements are colored in cyan). The SAVED domain of protomer B (and D) forms a "lid" of the pocket (secondary structure elements are colored in beige). The protein residues interacting with the 3′2′-c-GAMP ligand bound between the SAVED domains of dimer A/B (and C/D) are indicated by the red triangles.

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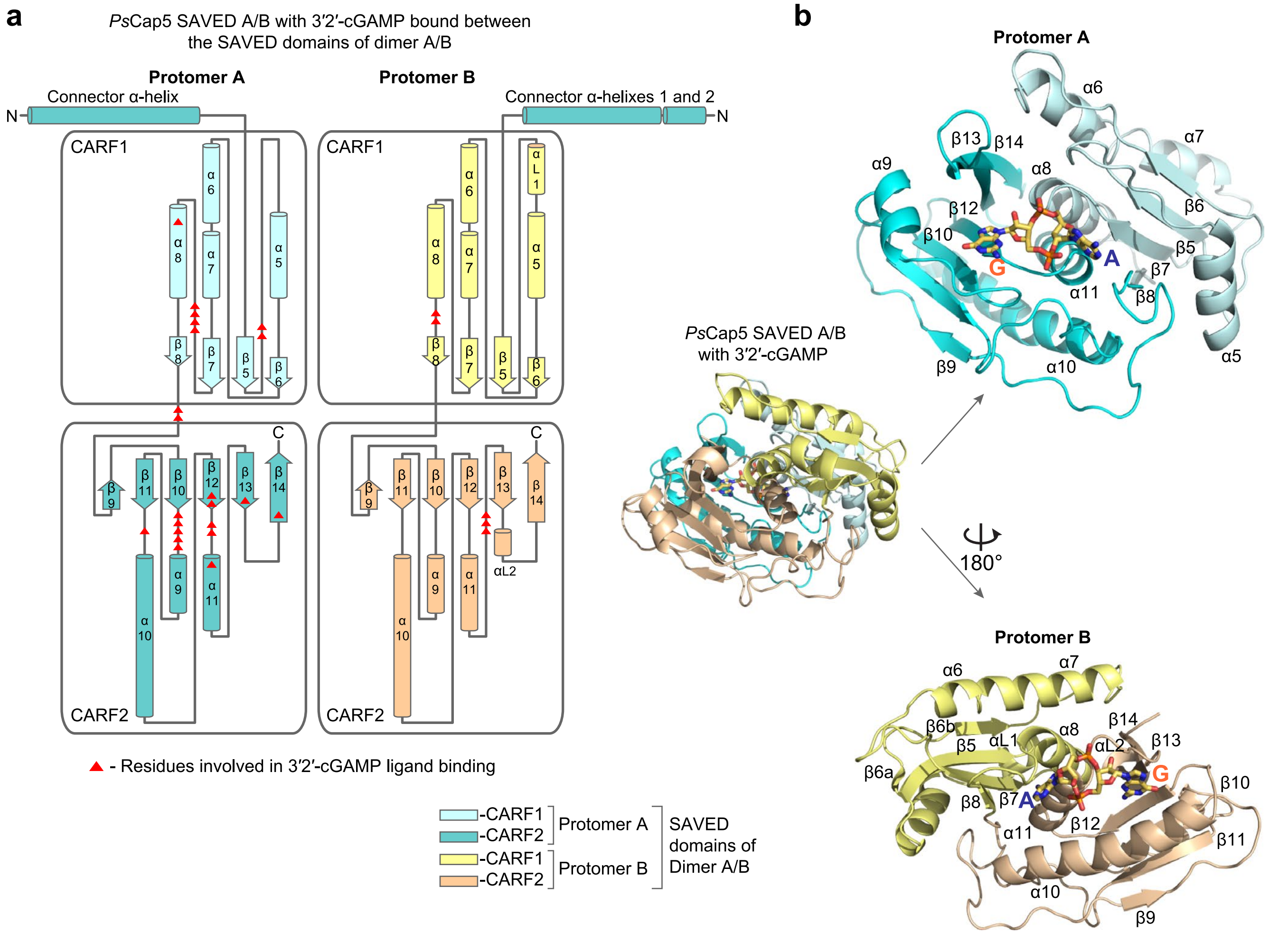
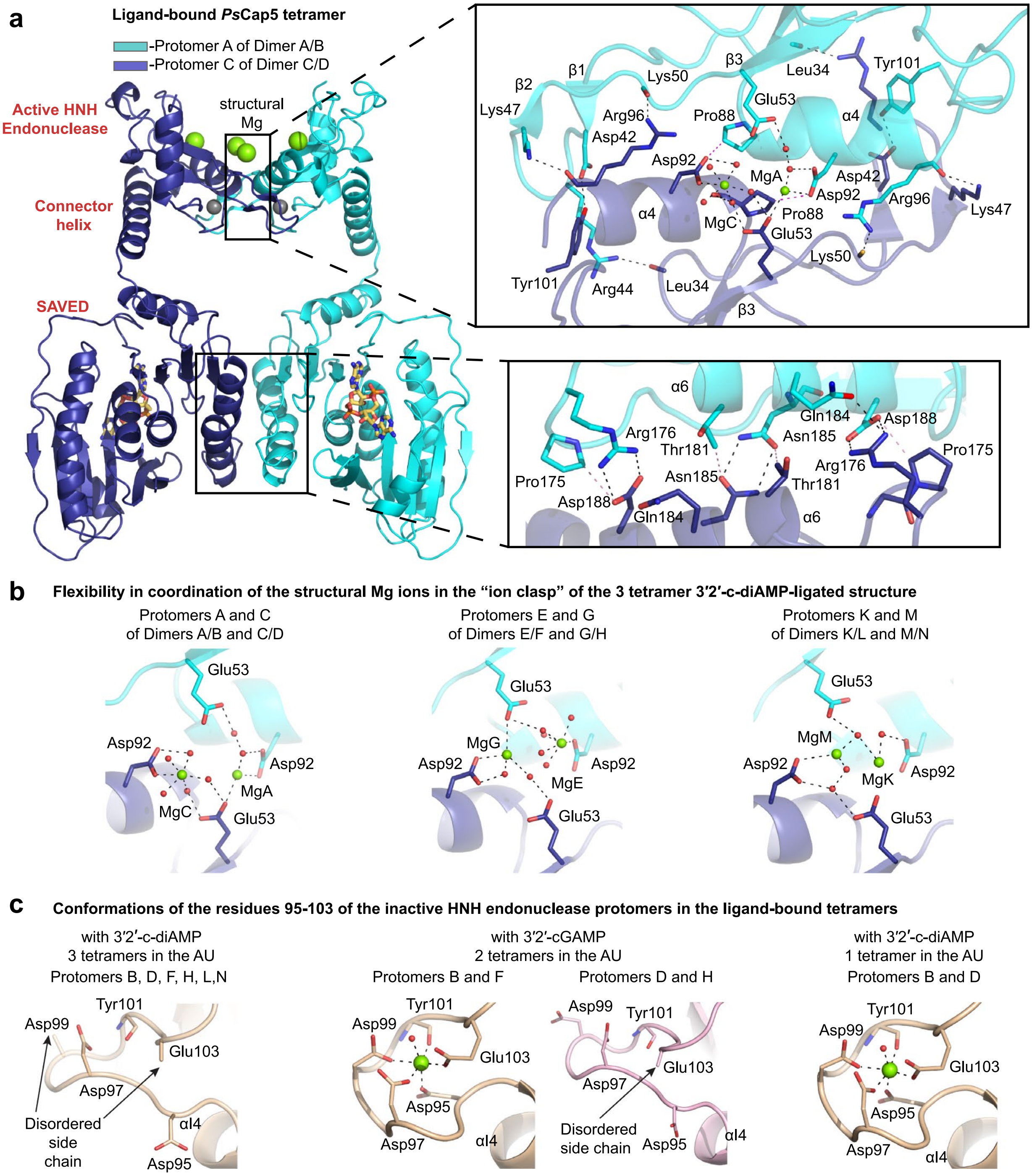


Extended Data Fig. 3 | Two CARF-like unit architecture of the SAVED domain of *Ps*Cap5 and ligand recognition. a, Topology diagram of the SAVED domain of *Ps*Cap5 highlights that the SAVED domain has a pseudo-symmetry and is a fusion of two different in sequence CRISP-associated Rossman fold (CARF) domains with alternating β-strands and α-helical segments. CARF1 modules of protomers A and B shown in light cyan and yellow, respectively, and CARF2 modules of protomers A and B are colored in cyan and light orange, respectively. In the *Ps*Cap5 tetramer, the activating ligand is bound between the SAVED domains of the crisscross dimer A/B (and C/D). The protomers A and B use different residues of CARF1 and CARF2 modules for ligand recognition (labeled as red triangles). **b**, Structures of the SAVED domains of dimer A/B with CARF1/CARF2 modules colored as described above bound to the 3′2′-cGAMP ligand.



Extended Data Fig. 4 | Structural features of the ligand-bound *Ps*Cap5 tetramer. a, The interfaces between the HNH-HNH and SAVED-SAVED domains of protomers A and C of the *Ps*Cap5 tetramer shown in Fig. 2a are outlined by rectangles. In addition to protein-protein contacts bridging the two catalytically active HNH endonuclease domains of protomers A and C, a structural Mg^{2+} in each domain is coordinated by Asp92 and Glu53 to form an "ion clasp". **b**, Flexibility in coordination of the structural Mg^{2+} ions of the "ion clasp". The two structural Mg^{2+} ions are coordinated slightly different in the three tetramer assemblies of the AU of the 1.79 Å resolution 3′2′-c-diAMP containing structure. **c**, Conformations of Asn95 and of the linker residues 96 -103 immediately adjacent to the C-terminus of the inactive HNH endonuclease domains in the ligand-bound *Ps*Cap5 tetramers. The side chains of Asn95 and of the linker residues Asp97, Asp99, Glu103 and the main chain carbonyl of Tyr101 engage a different Mg^{2+} ion in several protomers (B and F of the 3′2′-cGAMP-ligated structure and protomers B and D of the one tetramer 3′2′-c-diAMP structure, thus helping to keep Asp95 in the inactive state. The α-helix of the characteristic for HNH endonucleases ββα module is shorter in the inactive form (residues 88-93) versus the active form (residues 88-96) and excludes Asp95 that is involved in coordination of the catalytic Mg^{2+} ion in the active form.

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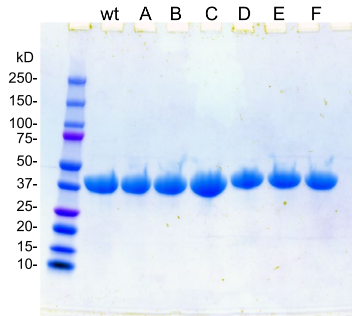
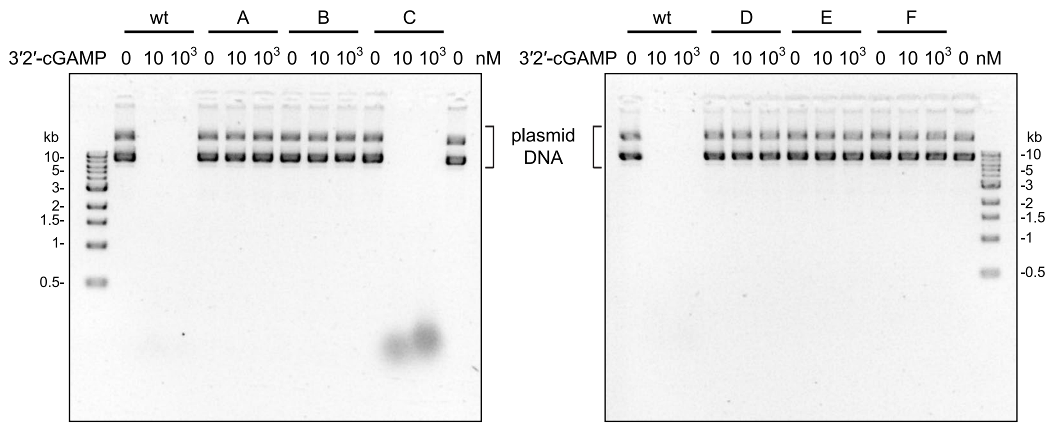
a Purified *wtPs*Cap5 and mutant proteins A-F



Code	*Ps*Cap5 mutants	Affected function in the ligand-bound tetramer
wt	wt *Ps*Cap5	Wild type
A	His56Ala	Catalytic residue
B	His91Ala-Asp95Ala	Catalytic ion binding
C	Glu53Ala-Asp92Ala	Structural ion binding and the HNH A/C domain interface
D	Asn185Trp-Asp188Arg	SAVED A/C domain interface
E	Asn185Arg-Asp188Arg	SAVED A/C domain interface
F	His138Ala-Arg242Ala-Ser277Ala	Binding of the phosphate groups of the cyclic dinucleotide ligand

b



Extended Data Fig. 5 | Plasmid DNA digestion by mutant *Ps*Cap5 proteins. **a**, Purity of wt*Ps*Cap5 and mutant *Ps*Cap5 proteins. Wt, wild type; A, His56Ala mutant; B, His91Ala-Asp95Ala double mutant; C, Glu53Ala-Asp92Ala double mutant; D, Asn185Trp-Asp188Arg double mutant; E, Asn185Arg-Asp188Arg double mutant, and F, His138Ala-Arg242Ala-Ser277Ala triple mutant. About 25 µg of each protein is loaded on the gel. The functional implications of each mutant are briefly listed in the table. **b**, Plasmid DNA digestion assay by wt*Ps*Cap5 and mutant *Ps*Cap5 proteins with 3′2′-cGAMP (3′2′GA) ligand. Concentrations of the *Ps*Cap5 proteins are 50 nM and the concentrations of 3′2′-cGAMP are 0, 10 and 1,000 nM as indicated. Wt*Ps*Cap5 digests plasmid DNA completely with 10 nM 3′2′-cGAMP. Mutants A, B, D, E, and F are inactive. The activity of mutant C is impaired since there is some undigested DNA products are left even with 1,000 nM activating ligand. The DNA cleavage products are analyzed by agarose gel electrophoresis followed by staining with ethidium bromide. The black and white colors are inverted for clarity. Plasmid DNA includes closed and open circle species. Each gel is a representative of at least three independent experiments.

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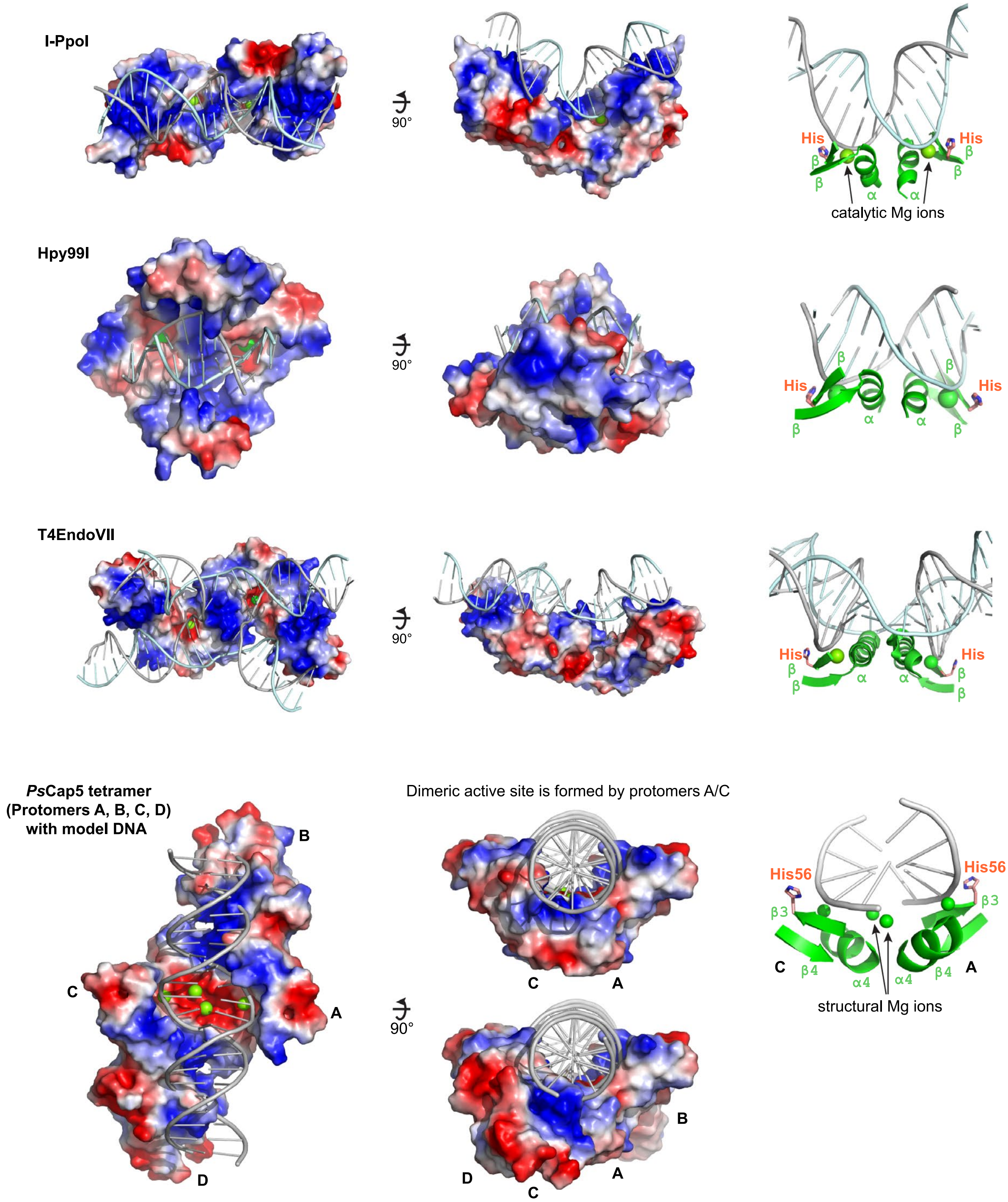

Extended Data Fig. 6 | Structures of dimeric HNH endonuclease complexes with DNA and their comparison with the dimeric HNH active site of *Ps*Cap5. HNH endonuclease I-PpoI[18] (PDB ID: 1A74, with the catalytic ions modeled based on the complex with cleaved DNA PDB ID: 1A73) and Hpy99I[19] (PDB ID: 3GOX). In the displayed Hpy99I structure, the N-terminal β-barrel (residues 1–53), a linker (residues 54–64) and the residues of the HNH domain which approach the major groove of the DNA for sequence-specific interactions (residues 79-98) have been omitted for clarity. The phage T4 Endo VII[20] (PDB ID: 2QNC) cleaves two symmetrical strands of a four-way junction DNA. For all proteins, a front and a side views of a DNA-binding surface colored by electrostatic potential are shown; the conserved ββα structural elements of the dimer approaching duplex DNA for cleavage from the minor groove side is colored in green, the catalytic ions are shown as green spheres and the catalytic His residues shown as red sticks. A front view of a DNA binding surface of dimeric HNH endonuclease of *Ps*Cap5 tetramer, formed by the active conformation protomers A and C. The view also includes the inactive protomers B and D. The side view of HNH endonuclease domains A/C only and of the four endonuclease domains of the tetramer. The two structural Mg^{2+} ions are positioned in-between of the catalytic ions. A model of HNH endonuclease domains A/C interactions with DNA is based on the structure of Colicin E9 HNH endonuclease monomer with DNA[22] (PDB ID: 1V15).

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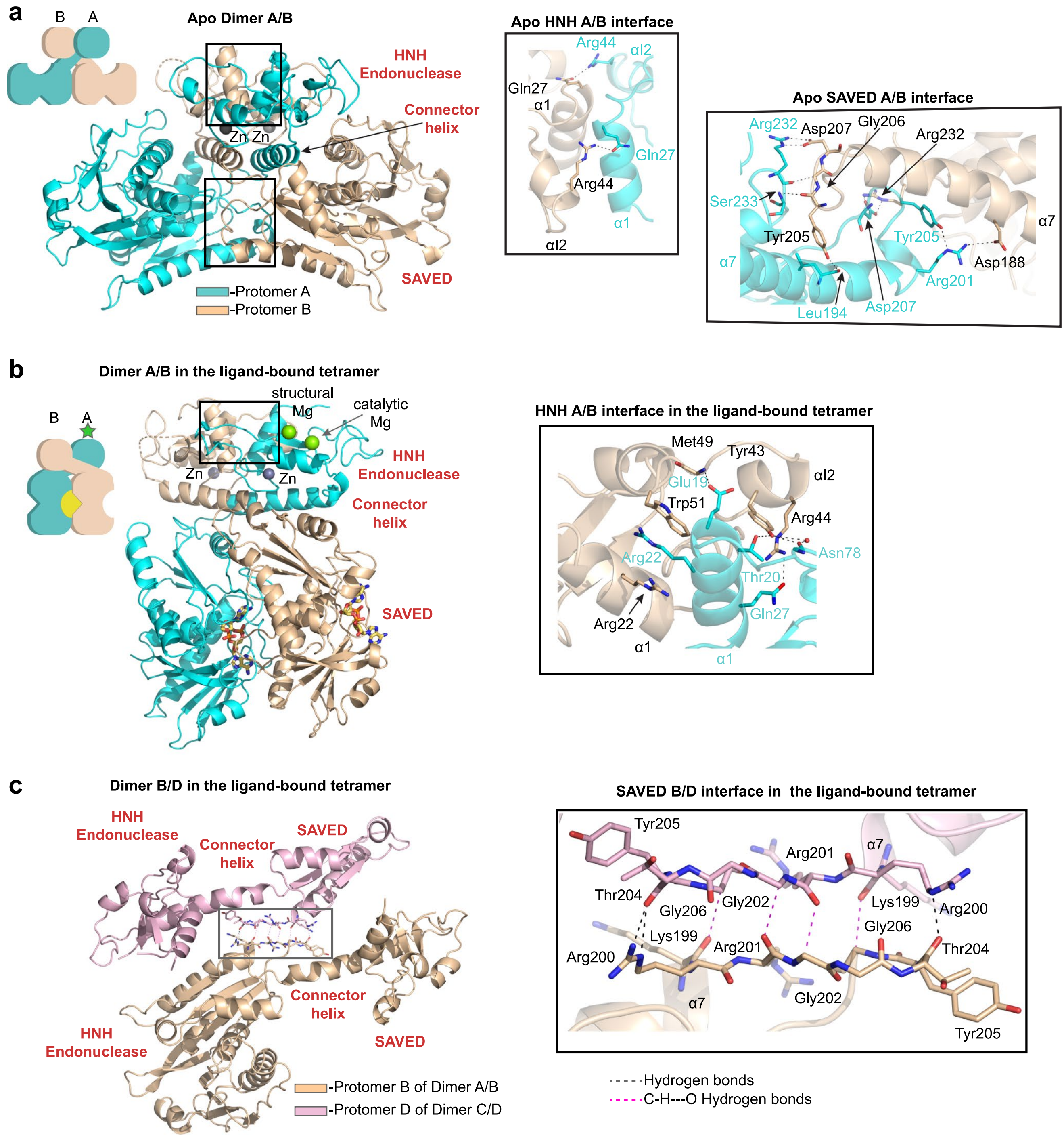

Extended Data Fig. 7 | Comparison of protein-protein interfaces in the apo dimer and in the ligand-bound *Ps*Cap5 tetramer. a, Overall structure of apo *Ps*Cap5 dimer with the interaction areas between the HNH endonuclease domains and the SAVED domains of protomers A and B highlighted by rectangles. The inserts show the details of the interactions between the HNH endonuclease domains and the SAVED domains. **b**, Overall structure of protomers A and B in the tetramer with the interaction areas between the HNH domains of the protomers indicated by black rectangle. The insert shows the details of the interactions. **c**, Overall structure of protomers B and D in the tetramer with the interaction area between the protomers indicated by black rectangle. The interactions between protomers B and D involve SAVED domain α7 residues 199-201 and adjacent loop residues 202-205 of both protomers. Interestingly, these interactions occur almost exclusively via weaker more transient C-H−O hydrogen bonds (pink dash lines) rather than conventional hydrogen bonds (black dash lines). Notably, in the apo dimer, the same residues remodel into a different conformation and provide the interface between SAVED A and B of the apo dimer via conventional hydrogen bonds (panel **a**).

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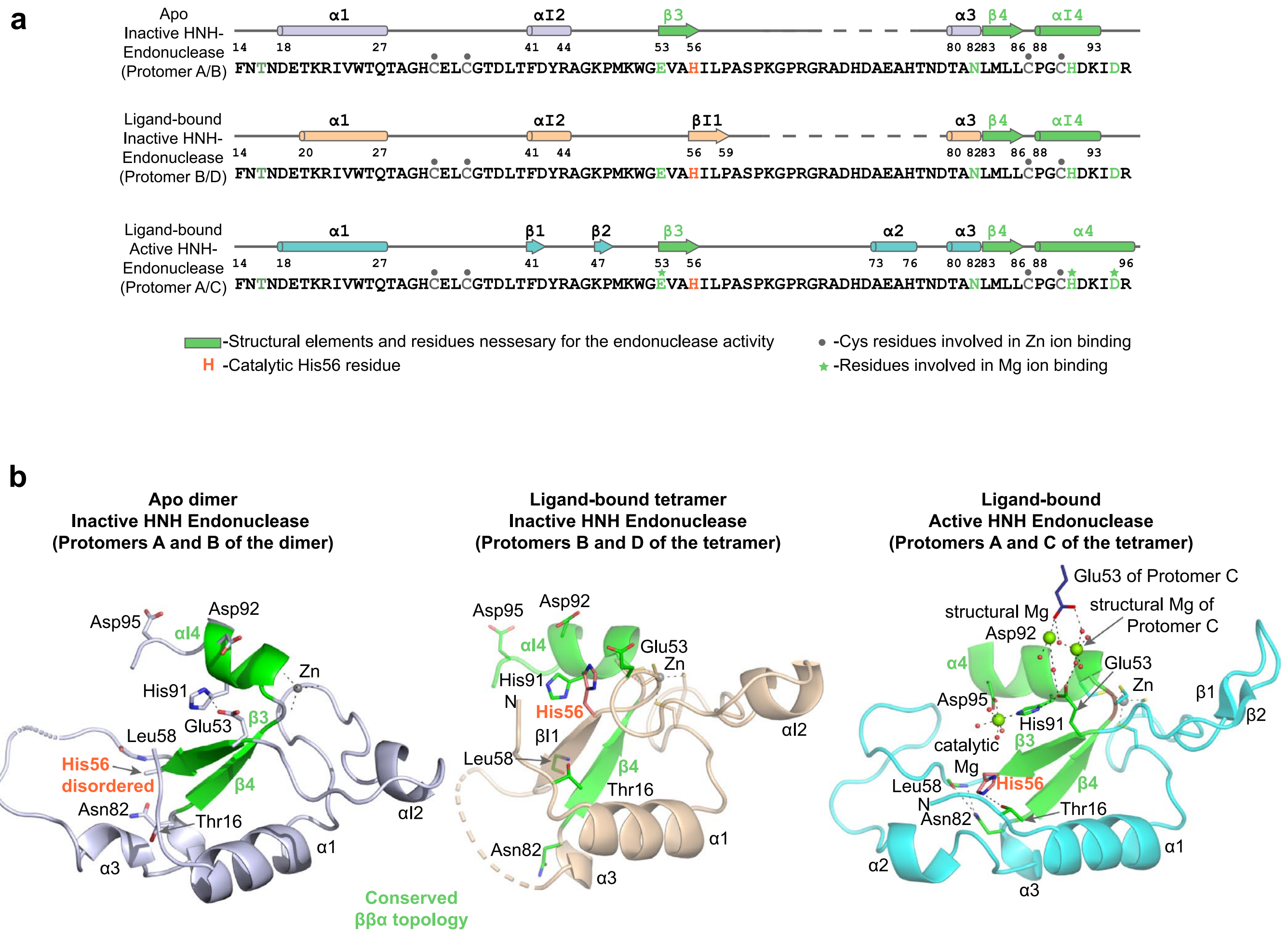

Extended Data Fig. 8 | Inactive conformation of the HNH endonuclease domain in the apo *Ps*Cap5 dimer and its comparison with the inactive and active conformations of the HNH endonucleases in the ligand-bound tetramer. a, Sequence and secondary structures of catalytically inactive endonucleases in the apo dimer A/B (A is simitar to B and colored light blue), and inactive and active conformations of the HNH endonuclease domains in the tetramer (beige and cyan). In the tetramer, HNH endonucleases of protomers A (and C) have active conformation with the characteristic ββα-topology of HNH endonucleases colored in green. Catalytic His56 is highlighted in red. Residues that are involved in coordination of the catalytic and structural Mg^{2+} in the catalytically active conformation are indicated by green stars. Cys32, Cys35, Cys87 and Cys90 coordinate a structural Zn^{2+} ion and are indicated by the gray dot. **b**, Structures of the HNH endonuclease domain in the inactive conformations in the apo dimer and ligand-bound tetramer and the catalytically active conformation in the tetramer. In the apo structure, the side chain of the catalytic His56 is disordered in both protomers in all six copies of the dimer assembly of the AU; the catalytic and structural Mg^{2+} are lacking. Furthermore, the α helix of the ββα-topology is shorter than in the active conformation (αI4, residues 88 - 93 vs α4, residues 88 - 96), this positions the side chain of Asp95 ~ 7 Å away from the putative catalytic Mg^{2+}-coordinating position. See the main text and Fig. 3 for description of the inactive and active conformations of HNH endonucleases in the tetramer.

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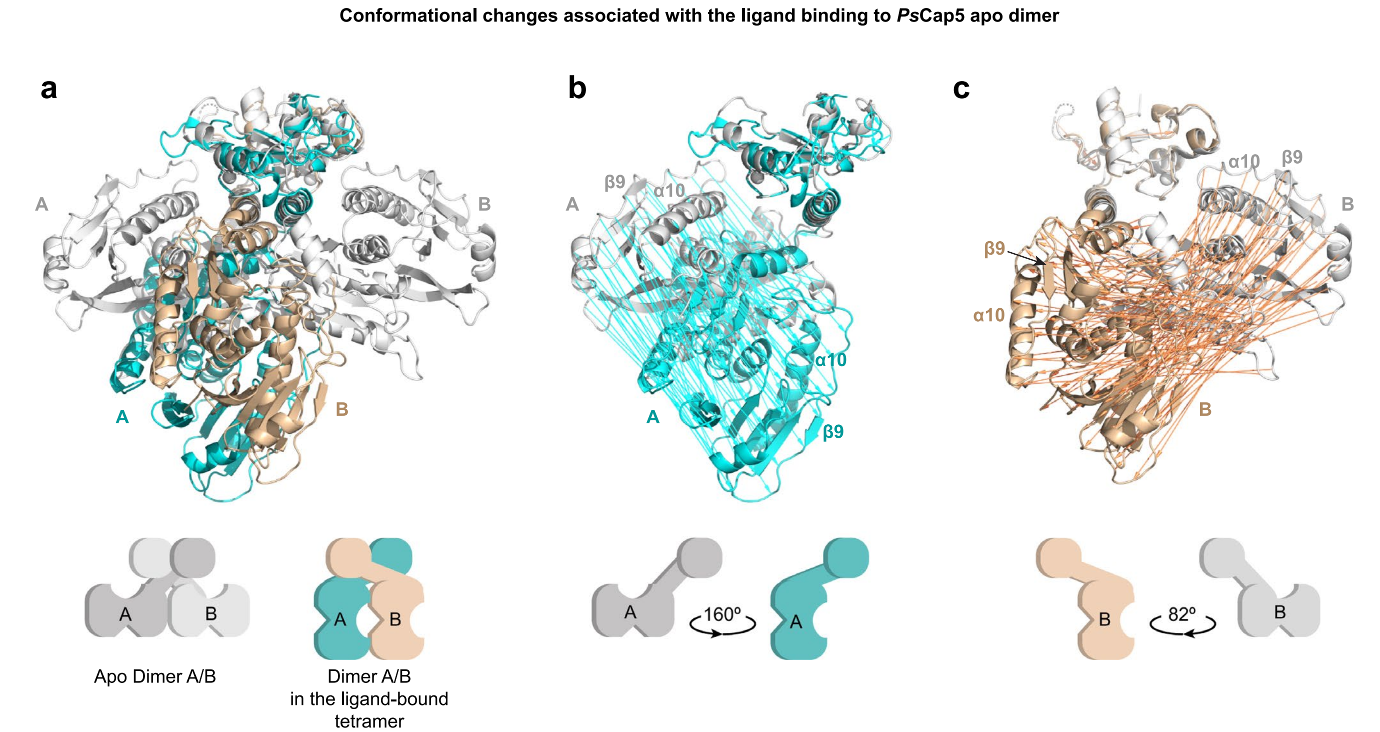

Extended Data Fig. 9 | Conformational changes associated with the ligand binding to apo *Ps*Cap5 dimer. a, Overlay of apo *Ps*Cap5 A/B dimer and dimer A/B in the ligand-bound activated *Ps*Cap5 tetramer. The structure of apo *Ps*Cap5 crisscross dimer with the SAVED domains in the "open" conformation is colored light gray. The structure of the ligand-bound dimer (protomers A and B; one of the two dimers in the tetramer) is colored in cyan for protomer A and beige for protomer B and shown in the same view as in Fig. 5. The structures are superimposed by the HNH endonuclease domains of protomers A and B. The SAVED of protomer A rotates by 160° and presents the opposite surface to the SAVED of protomer B with the β9-α10 structural element facing inward. Protomer B on the other hand rotates by 82° with the β9-α10 structural element facing outward. **b** and **c**, Vector map of global conformational changes upon the activating ligand binding to apo *Ps*Cap5 for protomers A and B, respectively.

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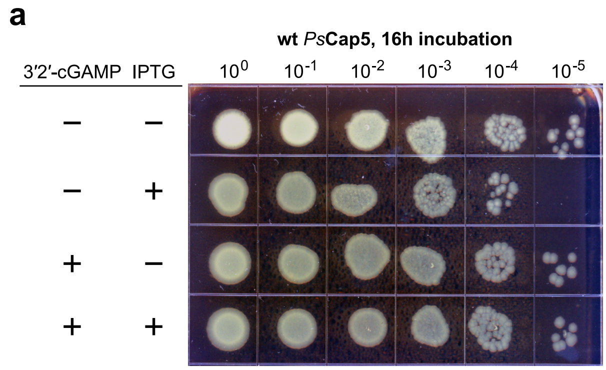

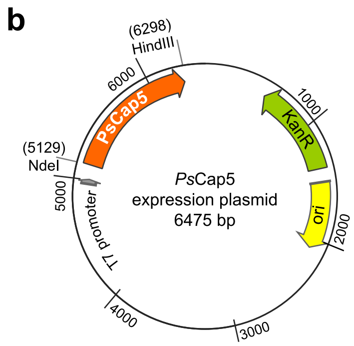

c

#	Plasmid size, bp	Excised region	Lost portion of *Ps*Cap5 gene
1	2,468	2449–21	all
2	3,587	3067–522	all
3	3,392	2585–5667	HNH
4	4,384	4658–273	all
5	2,544	3067–522	all
6	3,477	2401–5398	all
7	3,079	5666–2586	HNH
8	3,566	3614–47	all
9	2,941	3080–138	all

Extended Data Fig. 10 | Recovery of bacteria growth in the presence of extrinsically supplied 3′2′-cGAMP due to loss of the *Ps*Cap5 gene. a, Serial dilution of wt*Ps*Cap5 plasmid carrying *E. coli* culture in the absence or presence of IPTG and *Ps*Cap5-activating 3′2′-cGAMP ligand. After reaching OD_{600} ~ 0.6 the cultures were incubated for 16 h with or without IPTG and 3′2′-cGAMP, serially diluted and plated. In contrast to the pronounced reduction in bacterial growth observed after 3 h of incubation (Fig. 6a), the number of bacterial cells after longer 16 h incubation is not affected by an addition of 3′2′-cGAMP. We note high expression levels of *Ps*Cap5 protein either with or without IPTG induction (Fig. 6b). The plate is a representative of at least 3 biologically independent experiments. **b**, A map of the 6475 bp *Ps*Cap5 expression plasmid, where the DNA sequence coding for wt*Ps*Cap5 is inserted between NdeI and HindIII cloning sites of pET28b(+) vector. **c**, The loss of *Ps*Cap5-coding DNA sequence from plasmids isolated from the bacterial culture incubated with 3′2′-cGAMP and IPTG for 16 h. We have found 9 unique plasmids ranging in size from 2,468 to 4,384 base pairs (bp). We were not able to isolate any original 6475 bp *Ps*Cap5-carrying plasmids from the culture. DNA sequencing revealed the complete loss of the *Ps*Cap5-coding sequence in the deletion mutants #1, #2, #4, #5, #8 and #9. Mutants #3 and #7 have lost the coding sequence for the HNH endonuclease domain of *Ps*Cap5 and mutant #6 has lost the HNH and most of the SAVED domain. All plasmids retained kanamycin resistance gene since all the cultures were grown with kanamycin antibiotic.

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☒	☐	A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☒	☐	For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
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Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	X-ray diffraction data were collected using LSDC v2.0.2 computer software available at AMX 17-ID-1 beamline at NSLS-II at Brookhaven National Laboratory, NY and custom browser based general user interface (GUI) at NE-CAT 24-ID-E beamline at Advanced Photon Source (APS) at Argonne National Laboratory, IL.
Data analysis	X-ray diffraction data were processed with autoPROC version 1.0.5. Structure solution and refinement were carried out with PHENIX software suite version 1.20.1-4487. In Silico model for molecular replacement was generated by AlphaFold2 (AlphaFold2 advanced beta version) derived by the ColabFold server. For the apo PsCap5 structure refinement we also used AlphaFold-multimer ColabFold v1.5.3 prediction with the ColabFold server. Structure models were manipulated and electron density maps were visualized in Coot version 0.9.6. All molecular graphic figures were prepared using PyMOL version 2.5.5. The interaction diagrams for the protein contacts with the ligands was produced with the Maestro module version 11.8.012 in the Schrodinger suite of programs (Schrodinger LLC). Surface areas were calculated using PDBePISA version 1.52 (the protein interfaces, surfaces and assemblies service (PISA) at the European Bioinformatics Institute; http://www.ebi.ac.uk/pdbe/prot_int/pistart.html). The gels were photographed with GelDoc-It2 imaging system running VisionWorks version 8.17.16133.9147 software. The Refeyn mass photometry datasets were recorded with the AcquireMP version 2023 R1.1 and analyzed with the DiscoverMP version 2023 R1.2 software.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

Atomic coordinates and structure factors for PsCap5 with 3'2'-cGAMP, 3'2'-c-diAMP (with three and one tetramer assemblies in the AU) and without a ligand are available in the Protein Data Bank (PDB) under accession codes 8FMH (https://doi.org/10.2210/pdb8fmh/pdb), 8FMG (https://doi.org/10.2210/pdb8fmg/pdb), 8FMF (https://doi.org/10.2210/pdb8fmf/pdb), and 8FM1 (https://doi.org/10.2210/pdb8fm1/pdb), respectively. Associated data are provided in Extended Data Figures 1–10. The unprocessed images of the electrophoretic gels are supplied in Source Data. Structures with PDB IDs 1A74, 1A73, 3GOX, 2QNC, 1V15 were used in the study.

Human research participants

Policy information about studies involving human research participants and Sex and Gender in Research.

Reporting on sex and gender	N/A
Population characteristics	N/A
Recruitment	N/A
Ethics oversight	N/A

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Sample sizes of 151,489, 267,447, 64,174, and 85,838 X-ray reflections were sufficient to provide enough data to refine the structures of PsCap 5 with 3'2'-cGAMP, 3'2'-c-diAMP (with three and one tetramer assemblies in the AU) and without a ligand.
Data exclusions	No data were excluded during structure refinement using processed X-ray diffraction data sets.
Replication	All experiments carried in this study were reproducible. DNA plasmid digestion assays were repeated at least three times. Multiple X-ray diffraction data sets were collected with the best ones used for final refinement. 2 to 4 datasets for each structure were solved and found to be consistent with each other. For apo PsCap5, over 350 crystals were screened. For PsCap5 with 3'2'-cGAMP and 3'2'-c-diAMP, 32 crystals for each ligand have been screened.
Randomization	In each processed X-ray dataset, 5% of data were randomly selected and used during structure refinement for cross validation. Randomization is not relevant to the DNA plasmid digestion experiments because this study did not involve human subjects or animals.
Blinding	Blinding is not applicable to enzymatic and structural studies performed in this work because this study did not involve human subjects and animals..

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging